



PARHELION

A generator of trance

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About

Parhelion composes, mixes and plays trance: tracks of two to fourteen minutes and DJ sets of up to twelve hours, in which the tracks follow one another on two decks as a DJ would mix them. Nothing is played back from a recording and no neural network makes a sound. The kick, the sub and the rolling bass, the 303, the twelve lanes of the kit, the six voices of the supersaw synthesiser -- lead, counter, pluck, arp, pad and stab --, the physically modelled piano, the bowed strings, the choir, the brass and the timpani, the risers and impacts are synthesised; the rooms, the plate and the hall are simulated; and every section, every note and every move of a knob is decided by the composer.

Five style profiles span the genre: Uplifting (the anthem with its long breakdown, and, now and then, an orchestra -- Cinematic), Progressive (the plateau that rises in layers), Dream House (the piano motif over a bass that never stops), Acid (the 303 carrying bass and melody, its filter the drama) and Deep (the pad as the instrument, a grain cloud over it). Any two can be blended. Every recipe was fitted against measurements of thirty reference tracks: their tempo and length, the share and length of their breakdowns, how far the drop rises over the breakdown, their loudness and its range, their spectral balance and their width.

The same seed always gives the same track. A track is a seed, a style, a length and a handful of rerolls, which is what a set file stores and what you can hand to someone else. The standalone and the plugin play what the offline renderer plays, sample for sample, whatever the block size.

A parhelion is a sun dog: the bright spot beside the sun, on the halo of ice crystals, that shines with the sun's own light. In trance the drop is the sun; everything before it is the halo.

Quick start

The standalone plays by itself: press Play. Beside the key, choose what Parhelion writes: a single Track or a DJ mix, and its Length beside them (a track 2 to 14 minutes, a mix 10 minutes to 12 hours). A click on either composes it at once; Compose (it reads "Compose track" or "Compose mix") writes another with the knobs as they stand, and is lit while what plays is the other kind; New seed draws a fresh one. The status line begins with what plays -- "single track" or "DJ mix of 6 tracks" -- and tells its style and form, its tempo, its key with the Camelot label a DJ mixes by, and its length. While a track begins to play, Parhelion measures it in the background: each voice is set against the kick in a window of its own, every breakdown against the drop by the distance the style's references keep, every build a little under the drop, and the drop to the style's loudness. The status says "levelling" until the corrections have glided in.

A DJ mix (a set: the Set page names it so) is a dramaturgy of energy and tempo, tracks in neighbouring keys on the Camelot wheel, the incoming track's intro under the outgoing track's outro, the bass swapped on the mixer's isolator, the next track's hook teased on the third deck. Its first track may open on its atmosphere, as a track alone does -- nothing is mixed into it --; every other track opens with its DJ intro, the kick and the hats from the first bar.

In a DAW, Parhelion is an instrument plugin that follows the host's transport: it plays when the host plays, jumps when the host jumps, and plays at the host's tempo (a set's own tempo drift is the standalone's and the export's). Its output is a finished stereo mix, its loudest 20 seconds at the style's loudness (-8 to -10 LUFS, as the references) and its true peak under -1 dBTP.

A reroll draws one part of the track again and leaves the rest as it was: the form, the layer matrix, the energy, the harmony, the motif, the lead, the bass, the 303, the arp, the pluck, the piano, the orchestra, the drums, the effects, the sounds. In a set the rerolls act on the track under the playhead, and "reroll the whole track" or "reroll the mix's plan" draw more at once (the Arrange page).

The track and the set



A track has one of five forms, and each style draws its own. The Anthem is Uplifting's: intro, groove, a short break and a first drop, the main breakdown of 32 to 64 bars with its tease, its peak and its build, the main drop, and now and then a second break and a final drop, the outro. The Dream is Dream House's, shorter, and the bass never leaves, not even in the breakdown. The Acid form has no long breakdown: one to three pauses of the kick of eight bars each, between grooves, and the main drop; its drama is the 303's filter. The Plateau is Progressive's: grooves and plateaus that add a layer at a time, short breaks between them, a breakdown of 24 or 32 bars. The Drift is Deep's, a long groove and plateau before the breakdown and the main drop. Every section is a multiple of eight bars and the whole a multiple of thirty-two; every 8-bar block changes something against the one before it; the bar before a drop empties -- the last beat, the last two, the whole bar, or the kick alone on four -- so that the drop lands.

The intro opens on an atmosphere. The pad plays from the first bar, filtered dark, and opens over the intro; the effects lay a long atmosphere under it (a swell of detuned saws and noise in the key) with reverse swells into the 16-bar lines, and in the intro's second half the arp's pattern ghosts through, filtered. Under that the DJ intro builds as a DJ needs it: the kick, the hats, the bass, the open hat, the percussion. A track that stands alone -- not in a set -- may open without the kick altogether, on the pad and the atmosphere (Progressive, Dream House and Deep often, Uplifting and Acid now and then); a set's tracks always start with the kick, so they can be mixed. How much sustained sound an intro carries was measured on the references (the sustained energy between 200 Hz and 5 kHz in the first 16 and 32 bars against the drop) and the intros are held to it.

The harmony: a key the profile favours (the roots from D to A most often), one of five progressions (i-VI-III-VII and its relatives; the major keys through their relative minors), the harmonic rhythm of the section, the VII in

the two bars before every drop and its resolution on i. Acid stays on its chord and moves a fourth or a fifth every 16 bars; the Cinematic tracks lift by a third after the second breakdown.

The melody is written from rules and from counts. The anthem's motif is long notes: a leap of a fourth to an octave after the first note, then down by steps, a chord tone on every beat, phrased A A' B A''; filtered sections hear only A, the main drop's second half an octave higher. The riff, the pluck's rhythm, the piano's line and the 303's bar are drawn from note statistics of order two, never from a melody itself, and every lead is checked two bars at a time against a fingerprint of 144 250 windows of known trance tracks and drawn again on a match.

Style picks the profile; Morph To and Morph blend it with a second one. With Auto the tempo and the key are drawn from the profile; without it the knobs decide. Length is the track's, rounded to whole 32-bar blocks. Humanize is the spread of the velocities and the timing. Composer's Sounds lets the composer choose a preset per synth for every track (the next chapter); Favor Ratings weighs what you rated (the Arrange page).

A set is a length in minutes and a dramaturgy -- Warm-up, Peak, Closing, Sunrise (from the deep end up into the light) and Journey (a long arc with two peaks) --, which draws the energy and the tempo of every track (the tempo moves by at most two BPM a track). Styles: Stay keeps the style of the knobs, Wander walks the ladder Deep, Dream House, Progressive, Uplifting, Acid with the energy. The keys walk the Camelot wheel: the same key, a fifth, the relative major or minor -- from major back to minor more often, trance lives in minor --, now and then an energy boost by one or two semitones. Every track of a set can be mixed: an intro and an outro of 32 bars or more, with a beat, never beatless.

The blend lays the incoming track's intro over the outgoing track's outro, 32 or 64 bars (Blend). The bass goes over on the isolator in one move, 16 bars before the incoming groove, never before the incoming bass; the DJ's hand works the faders and the filters. On deck C the next track's hook is teased, filtered, over the end of the one playing. Track Time is how long each track is heard on its own. The mixer's effects (djfx) are an echo and a hall that the breaks and the Perform page's throw send into.

COMPOSE

Control	Key	Range	Default
Tempo	<code>compose.bpm</code>	90 .. 150 BPM	138 BPM
Key	<code>compose.key</code>	C, C#, D, D#, E, F, F#, G, G#, A, A#, B	A
Scale	<code>compose.scale</code>	Aeolian, Dorian, Harmonic Minor, Minor Pentatonic, Phrygian, Ionian	Aeolian
Style	<code>compose.style</code>	Uplifting, Progressive, Dream House, Acid, Deep	Uplifting
Length	<code>compose.minutes</code>	2 .. 14 min	8 min
Auto	<code>compose.auto</code>	off / on	on
Morph To	<code>compose.morph_to</code>	Off, Uplifting, Progressive, Dream House, Acid, Deep	Off
Morph	<code>compose.morph</code>	0 .. 1	0.5
Composer's Sounds	<code>compose.pick_sounds</code>	off / on	on
Favor Ratings	<code>compose.use_ratings</code>	off / on	off
Humanize	<code>compose.humanize</code>	0 .. 30 %	8 %

SET

Control	Key	Range	Default
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Dramaturgy	set.dramaturgy	Warm-up, Peak, Closing, Sunrise, Journey	Peak
Styles	set.journey	Stay, Wander	Stay
Blend	set.blend	32 bars, 64 bars	32 bars
Set Length	set.minutes	0 .. 720 min	0 min
Track Time	set.track_minutes	3 .. 10 min	6 min

DJFX

Control	Key	Range	Default
Echo Time	djfx.echo_time	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/8
Feedback	djfx.feedback	0 .. 1	0.45
Echo Return	djfx.echo_return	-60 .. 6 dB	-3 dB
Hall Decay	djfx.hall_decay	0.5 .. 12 s	5 s
Hall Return	djfx.hall_return	-60 .. 12 dB	6 dB

Sounds and presets

Every synth has 1024 factory presets: the kick, the sub, the kit (by role), the bass, the 303, each of the six voices of the supersaw synthesiser (lead, counter, pluck, arp, pad, stab), the piano, the strings, the choir, the brass, the timpani, the effects and the grain cloud -- 18 banks, 18 432 sounds. Each bank has sixteen groups -- "Anthem Supersaw", "Classic 303", "Gated Pad", "Concert Grand" -- and each group sixty-four presets on an eight by eight grid: the rows an adjective from dark to bright, the columns a noun of the group's own, so the name tells where on the grid a sound lies. A preset is the sound with its modulation, not the mix: levels, sends, the pump and the notes stay where the mix and the composer put them. Each preset's loudness against the default sound was measured, and the composer trims it by that amount.

For every track the composer chooses a preset per synth (Composer's Sounds, `compose.pick_sounds`): a group by how well it suits the track's style -- the weights cubed, so that an Uplifting lead is mostly a supersaw --, a lane of the kit only among the groups made for its role, and in the group a row around the style's brightness (Uplifting brighter, Deep darker) and a column at random. The style then holds a few knobs in its own range. Reroll sounds draws other presets and leaves the notes as they are.

The knobs show what plays. When a track begins -- in a set, when the next track starts on its deck -- its presets are put on the knobs, and each voice's preset bar names the preset the composer chose; the Arrange page lists them for the whole track. Turn a knob and the sound follows from there; choose another preset from the menu, or step through them with the arrows, and it plays at once. In a blend each deck keeps its own track's sounds.

Save as user preset (the last item of the menu) keeps a sound of your own: every knob a preset sets, as it stands, goes under a name into Documents/Parhelion/Presets/<synth>, one "knob=value" a line, and appears in the menu under User. The twelve lanes of the kit share their folder, so a lane's sound loads into any lane. A user preset is the whole sound, as a factory preset is: a knob its file does not name (one edited out by hand) goes back to its default.

Every voice has the same modulation block: a modulation envelope, four LFOs (free or in time with the beat) and a matrix of eight slots, each a source, a destination and an amount. The sources: the envelope, the LFOs, velocity, the key, a random value per note, the mod wheel, channel pressure and the energy of the score, so that a voice can open with the track (the synthesisers add their filter envelope). The destinations are the engine's own: pitch, pulse width, wavetable position, FM index, cutoff, resonance, drive and detune for the synthesisers; hammer hardness for the piano; bow pressure, speed and position for the strings; vowel, formant, tension and breath for the choir; breath and brassiness for the brass; hardness, strike position and decay for the timpani; vibrato, level and pan for all of them. Most presets use a few slots; what the mod wheel does is the preset's.

Arrange

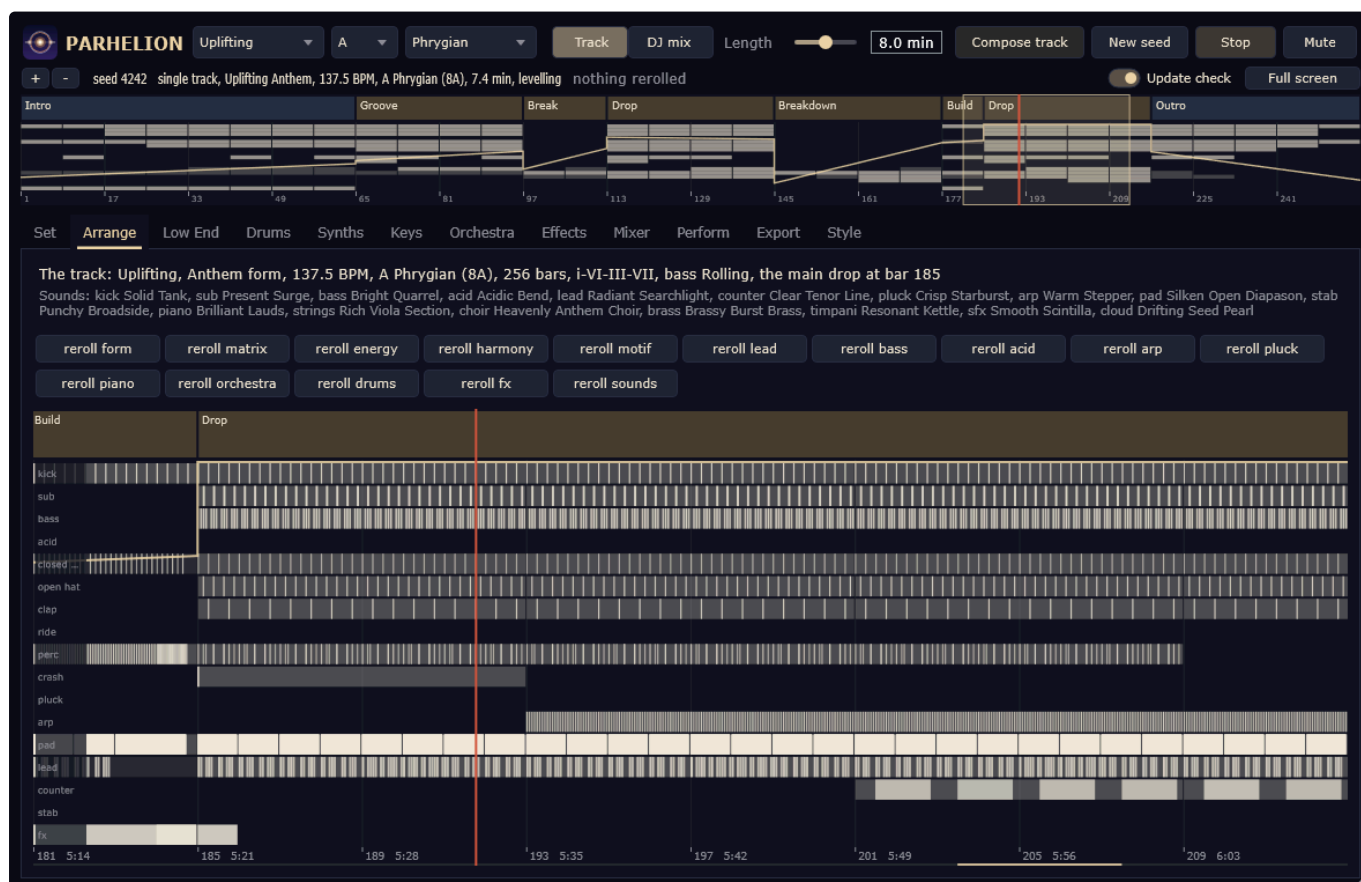


The arrangement across the whole length: in a track its sections by name, and under them a lane per layer -- kick, sub, bass, 303, the hats, the clap, the ride, the percussion, the crash, pluck, arp, pad, lead, counter, stab, effects --, bright where the layer plays open, dim where it plays filtered. Over them the energy curve the composer wrote. In a mix the tracks lie on decks A and B, their bass swaps marked, and the teasers on deck C. Under it a ruler: a track's bars and their time, a mix's minutes. A click jumps there.

The mouse wheel zooms in around the pointer, down to four bars; drag, or turn the wheel with Shift held (or sideways), to move along; a double click shows the whole length again. Zoomed in, the notes themselves stand in the lanes -- every kick, every hat, the chords of the pad, the lead's line --, a bar under the ruler shows where the window lies, and the strip on top of the panel marks it too; when the playhead runs out of the window, the window pages on with it (not when you moved it away). The same works on the strip on top.



A DJ mix of forty minutes: its tracks on decks A and B, the ruler in minutes



Zoomed in on a main drop: the notes in the lanes, the window marked on the strip on top

The line above names the style, the form, the tempo, the key, the progression, the bass figure, the main drop and, where there is one, the orchestra and the key change; the line below the preset of every synth.

The row of rerolls acts on the track under the playhead; each part is drawn from a stream of its own, so a reroll of the matrix keeps the harmony and the sounds, a reroll of the energy redraws the curve the pads' opening and the 303 follow, and a reroll of the motif writes another hook for the same form.

The + and - beside the status line rate the track under the playhead. The ratings are kept in Documents/Parhelion/ratings.tsv with the track's form and the groups of its presets; with Favor Ratings (the Set page) the composer draws the forms and preset groups you liked more often and the ones you did not seldom -- none is ever ruled out. A saved set does not carry the ratings: it plays back the same only with Favor Ratings off.

Low End



The low end is one system. The kick is tuned to the key (or its fifth, or its flat seventh); it has three engines (a swept sine, a resonator, a modelled 909) and a top layer, and its drive and clip run at four times the rate. The sub bass is a sine under the bass line, ducked by every kick and, with Kick Lock, locked to its phase, so that the two never cancel each other.

The mid bass and the 303 share one synthesiser with ten filter models (the Moog ladder, the 303's diode ladder, the Prophet's SSM2040, the Juno's IR3109, the SEM, the MS-20's Korg35 and more). The bass plays one of the trance figures -- the off-beat eighths, the rolling sixteenths, the gallop, the walking line, the drone --; the 303 plays its bar with accents and slides, and in Acid its cutoff, resonance, envelope and decay climb over phrases of 16, 32 or 64 bars and fall back at their end.

The pump is trance's sidechain: a ghost kick, silent and on a track of its own, ducks each voice by its own depth in dB, with the attack, hold and release here, and the returns of the rooms by Return Duck. It gives the pads and the leads their breathing without a compressor; in a breakdown the composer lets it go and ramps it back in for the drop.

KICK

Control	Key	Range	Default
Engine	<code>kick.engine</code>	Sweep, Resonator, 909	Sweep
Tune	<code>kick.tune</code>	Free, Key, Fifth, Flat Seventh	Key
Pitch End	<code>kick.pitch_end</code>	30 .. 120 Hz	52 Hz
Pitch Start	<code>kick.pitch_start</code>	60 .. 1500 Hz	220 Hz
Body Decay	<code>kick.pitch_decay</code>	5 .. 300 ms	24 ms
Punch Decay	<code>kick.punch_decay</code>	0.5 .. 20 ms	3 ms
Punch	<code>kick.punch</code>	0 .. 1	0.55
Attack	<code>kick.amp_attack</code>	0 .. 10 ms	0.2 ms
Hold	<code>kick.amp_hold</code>	0 .. 150 ms	20 ms
Decay	<code>kick.amp_decay</code>	20 .. 1500 ms	320 ms
Drive	<code>kick.drive</code>	0 .. 1	0.3
Clip	<code>kick.clip</code>	Soft, Hard	Soft
Click	<code>kick.click_level</code>	0 .. 1	0.4
Click Tone	<code>kick.click_tone</code>	500 .. 12000 Hz	3500 Hz
Click Decay	<code>kick.click_decay</code>	0.5 .. 30 ms	5 ms
Tone	<code>kick.tone</code>	200 .. 20000 Hz	8000 Hz
Level	<code>kick.level</code>	-36 .. 6 dB	-3 dB
Tail Limit	<code>kick.tail_limit</code>	-60 .. 0 dB	0 dB
Low Cut	<code>kick.low_cut</code>	15 .. 60 Hz	30 Hz
Dip Freq	<code>kick.dip_freq</code>	200 .. 1500 Hz	400 Hz
Dip	<code>kick.dip</code>	-12 .. 0 dB	-2 dB
Top Level	<code>kick.top_level</code>	-60 .. 0 dB	-10 dB
Top Pitch	<code>kick.top_pitch</code>	-24 .. 0 st	-5 st
Top Decay	<code>kick.top_decay</code>	20 .. 400 ms	70 ms

Top Drive	kick.top_drive	0 .. 1	0.4
Top Cut	kick.top_cut	80 .. 2000 Hz	100 Hz

SUB

Control	Key	Range	Default
Level	sub.level	-60 .. 6 dB	-4 dB
Octave	sub.octave	-2 .. 2	0
Attack	sub.attack	0.5 .. 50 ms	1 ms
Decay	sub.decay	20 .. 2000 ms	60 ms
Sustain	sub.sustain	0 .. 1	0
Release	sub.release	10 .. 800 ms	40 ms
Low Pass	sub.low_pass	40 .. 300 Hz	110 Hz
Drive	sub.drive	0 .. 1	0.1
Kick Lock	sub.lock	Off, Kick	Off
Duck	sub.duck	0 .. 24 dB	11 dB
Duck Hold	sub.duck_hold	0 .. 250 ms	20 ms
Duck Release	sub.duck_release	30 .. 600 ms	110 ms

BASS

Control	Key	Range	Default
Level	bass.level	-60 .. 6 dB	-1 dB
Pan	bass.pan	-1 .. 1	0
Wave	bass.wave	0 .. 1	0
Pulse Width	bass.pulse_width	0.05 .. 0.95	0.5
Sub Osc	bass.sub_osc	0 .. 1	0
Filter	bass.filter	Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, Wasp, Comb	Moog Ladder
Cutoff	bass.cutoff	40 .. 12000 Hz	420 Hz
Resonance	bass.resonance	0 .. 1	0.15
Env Amount	bass.env_amount	0 .. 6 oct	2.2 oct
Decay	bass.decay	5 .. 3000 ms	90 ms
Accent	bass.accent	0 .. 1	0.2
Attack	bass.amp_attack	0.5 .. 500 ms	0.5 ms
Amp Decay	bass.amp_decay	5 .. 3000 ms	150 ms
Sustain	bass.amp_sustain	0 .. 1	0
Release	bass.amp_release	5 .. 3000 ms	40 ms
Glide	bass.glide	0 .. 500 ms	0 ms
Drive	bass.drive	0 .. 1	0.25
Key Track	bass.key_track	0 .. 1	0.5
Low Cut	bass.low_cut	20 .. 500 Hz	80 Hz

High Cut	bass.high_cut	500 .. 20000 Hz	6000 Hz
Plate Send	bass.plate_send	0 .. 1	0
Room Send	bass.room_send	0 .. 1	0
Duck	bass.duck	0 .. 24 dB	8 dB
Duck Release	bass.duck_release	30 .. 600 ms	110 ms
Mod Attack	bass.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	bass.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	bass.menv_sustain	0 .. 1	0
Mod Release	bass.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	bass.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	bass.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	bass.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	bass.lfo1_retrigger	off / on	off
LFO 1 Fade	bass.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	bass.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	bass.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	bass.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	bass.lfo2_retrigger	off / on	off
LFO 2 Fade	bass.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	bass.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	bass.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	bass.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	bass.lfo3_retrigger	off / on	off
LFO 3 Fade	bass.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	bass.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	bass.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	bass.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	bass.lfo4_retrigger	off / on	off
LFO 4 Fade	bass.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	bass.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	bass.mx1_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 1 Amount	bass.mx1_amount	-1 .. 1	0
Mod 2 Source	bass.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	bass.mx2_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 2 Amount	bass.mx2_amount	-1 .. 1	0

Mod 3 Source	bass.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	bass.mx3_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 3 Amount	bass.mx3_amount	-1 .. 1	0
Mod 4 Source	bass.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	bass.mx4_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 4 Amount	bass.mx4_amount	-1 .. 1	0
Mod 5 Source	bass.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	bass.mx5_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 5 Amount	bass.mx5_amount	-1 .. 1	0
Mod 6 Source	bass.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	bass.mx6_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 6 Amount	bass.mx6_amount	-1 .. 1	0
Mod 7 Source	bass.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	bass.mx7_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 7 Amount	bass.mx7_amount	-1 .. 1	0
Mod 8 Source	bass.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	bass.mx8_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 8 Amount	bass.mx8_amount	-1 .. 1	0

ACID

Control	Key	Range	Default
Level	acid.level	-60 .. 6 dB	-12 dB
Pan	acid.pan	-1 .. 1	0.1
Wave	acid.wave	0 .. 1	0
Pulse Width	acid.pulse_width	0.05 .. 0.95	0.5
Sub Osc	acid.sub_osc	0 .. 1	0

Filter	acid.filter	Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, Wasp, Comb	Diode Ladder (303)
Cutoff	acid.cutoff	40 .. 12000 Hz	600 Hz
Resonance	acid.resonance	0 .. 1	0.72
Env Amount	acid.env_amount	0 .. 6 oct	2.5 oct
Decay	acid.decay	5 .. 3000 ms	220 ms
Accent	acid.accent	0 .. 1	0.6
Attack	acid.amp_attack	0.5 .. 500 ms	0.5 ms
Amp Decay	acid.amp_decay	5 .. 3000 ms	900 ms
Sustain	acid.amp_sustain	0 .. 1	0.8
Release	acid.amp_release	5 .. 3000 ms	60 ms
Glide	acid.glide	0 .. 500 ms	60 ms
Drive	acid.drive	0 .. 1	0.6
Key Track	acid.key_track	0 .. 1	0.3
Low Cut	acid.low_cut	20 .. 500 Hz	120 Hz
High Cut	acid.high_cut	500 .. 20000 Hz	9000 Hz
Plate Send	acid.plate_send	0 .. 1	0.1
Room Send	acid.room_send	0 .. 1	0.05
Duck	acid.duck	0 .. 24 dB	6 dB
Duck Release	acid.duck_release	30 .. 600 ms	110 ms
Mod Attack	acid.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	acid.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	acid.menv_sustain	0 .. 1	0
Mod Release	acid.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	acid.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	acid.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	acid.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	acid.lfo1_retrig	off / on	off
LFO 1 Fade	acid.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	acid.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	acid.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	acid.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	acid.lfo2_retrig	off / on	off
LFO 2 Fade	acid.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	acid.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	acid.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	acid.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	acid.lfo3_retrig	off / on	off
LFO 3 Fade	acid.lfo3_fade	0 .. 8 s	0 s

LFO 4 Rate	acid.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	acid.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	acid.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	acid.lfo4_retrigger	off / on	off
LFO 4 Fade	acid.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	acid.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	acid.mx1_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 1 Amount	acid.mx1_amount	-1 .. 1	0
Mod 2 Source	acid.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	acid.mx2_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 2 Amount	acid.mx2_amount	-1 .. 1	0
Mod 3 Source	acid.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	acid.mx3_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 3 Amount	acid.mx3_amount	-1 .. 1	0
Mod 4 Source	acid.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	acid.mx4_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 4 Amount	acid.mx4_amount	-1 .. 1	0
Mod 5 Source	acid.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	acid.mx5_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 5 Amount	acid.mx5_amount	-1 .. 1	0
Mod 6 Source	acid.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	acid.mx6_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 6 Amount	acid.mx6_amount	-1 .. 1	0
Mod 7 Source	acid.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	acid.mx7_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off

Mod 7 Amount	acid.mx7_amount	-1 .. 1	0
Mod 8 Source	acid.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	acid.mx8_dst	Off, Pitch, Pulse Width, Cutoff, Resonance, Env Amount, Drive, Level, Pan	Off
Mod 8 Amount	acid.mx8_amount	-1 .. 1	0

PUMP

Control	Key	Range	Default
Attack	pump.attack	0.1 .. 20 ms	0.5 ms
Hold	pump.hold	0 .. 150 ms	20 ms
Release	pump.release	30 .. 600 ms	120 ms
Return Duck	pump.return_duck	0 .. 12 dB	4 dB

Drums



Twelve lanes, each a small drum synthesiser: a role (the closed, rolling and open hats, the ride, the clap and its ghost, the snare, the rim, the shaker, toms, congas, noise, the crash, the tambourine), an engine (noise, metal - the 909's six square waves --, modal, tone, FM), a filter and a shape. They run side by side in the lanes of the processor's vector registers. The snare plays the roll into the drop, from quarters to thirty-seconds; the crash marks the drops.

The mix module holds the buses: the hats and the percussion with their level and low pass, the drum bus's saturation, the group low cut the builds draw up, and the synth bus's level -- the fader the energy curve moves.

PERC (12 INSTANCES)

Control	Key	Range	Default
Active	percN.active	off / on	on
Role	percN.role	Closed Hat, Rolling Hat, Open Hat, Ride, Clap, Clap Ghost, Snare, Rim, Shaker, Tom, Conga, Noise, Crash, Tambourine	Closed Hat

Engine	percN.engine	Noise, Metal, Modal, Tone, FM	Noise
Pitch	percN.pitch	40 .. 12000 Hz	400 Hz
Pitch Amount	percN.pitch_amount	1 .. 16 x	1 x
Pitch Decay	percN.pitch_decay	0.5 .. 300 ms	10 ms
FM Ratio	percN.fm_ratio	0.25 .. 8	1.41
FM Index	percN.fm_index	0 .. 8	0
Modes	percN.mode_set	Membrane, Bar, Harmonic	Membrane
Mode Damping	percN.mode_damp	0 .. 1	0.5
Metal Scale	percN.metal_scale	0.25 .. 4 x	1 x
Noise	percN.noise	0 .. 1	1
Noise Decay	percN.noise_decay	2 .. 3000 ms	55 ms
Bursts	percN.bursts	1 .. 6	1
Burst Spacing	percN.burst_spacing	2 .. 40 ms	10 ms
Decay	percN.decay	2 .. 3000 ms	55 ms
Filter	percN.filter	Low Pass, Band Pass, High Pass	High Pass
Cutoff	percN.cutoff	100 .. 18000 Hz	7500 Hz
Resonance	percN.resonance	0 .. 1	0.15
Low Cut	percN.low_cut	150 .. 8000 Hz	3000 Hz
Drive	percN.drive	0 .. 1	0.1
Level	percN.level	-36 .. 6 dB	-2.5 dB
Pan	percN.pan	-1 .. 1	0.1
Choke Group	percN.choke	0 .. 4	1
Shift	percN.shift	-10 .. 10 ms	0 ms
Density	percN.density	0 .. 1	0.5
Tune to Key	percN.tune	off / on	off
Pan Depth	percN.pan_depth	0 .. 1	0
Pan Period	percN.pan_bars	0.0625 .. 16 bars	0.1875 bars
Cut Tracks Pitch	percN.cut_track	0 .. 2	0
Noise Type	percN.noise_type	White, 909 Metal	909 Metal

MIX

Control	Key	Range	Default
Hats Level	mix.hats_level	-24 .. 12 dB	-1 dB
Hats Cut	mix.hats_cut	200 .. 20000 Hz	20000 Hz
Perc Level	mix.perc_level	-24 .. 12 dB	-4 dB
Perc Cut	mix.perc_cut	200 .. 20000 Hz	20000 Hz
Drum Saturation	mix.drum_sat	0 .. 1	0.2

Group Low Cut	<code>mix.low_cut</code>	20 .. 500 Hz	20 Hz
Synth Level	<code>mix.synth_level</code>	-24 .. 6 dB	0 dB



Six voices of one synthesiser, each with its own role and bank: the lead, the counter-melody, the pluck, the arp, the pad and the stab. Two oscillators -- the supersaw after Adam Szabo's measurements of the Roland JP-8000 (seven saws, the detune curve and the mix of the original), virtual analogue, FM or a wavetable --, a filter from the same ten models, an amplitude and a filter envelope, the trance gate (the classic patterns and their sixteenth-note masks), a tempo delay, sends into the room, the plate and the hall, and the modulation block. The page shows one voice at a time; the voice buttons switch between them.

The lead plays the motif; the counter answers in the lead's held notes an octave above; the pluck plays the chords in syncopated sixteenths; the arp breaks them up; the pad holds them, gated in the drops and opening over the breakdown; the stab hits them in the drops (every other block of the first drop, the main drop from its second block on). Which voices play in which block is the layer matrix's.

LEAD

Control	Key	Range	Default
Oscillator	lead.osc	Supersaw, VA, FM, Wavetable	Supersaw
Detune	lead.detune	0 .. 1	0.62
Mix	lead.mix	0 .. 1	0.8
Dynamic Detune	lead.dynamic_detune	0 .. 1	0.5
Wave	lead.wave	0 .. 1	0
Pulse Width	lead.pulse_width	0.05 .. 0.95	0.5
FM Ratio	lead.fm_ratio	0.5 .. 8	2
FM Index	lead.fm_index	0 .. 10	2.5
FM Decay	lead.fm_decay	5 .. 2000 ms	250 ms
Table	lead.table	Classic, Vocal, Glass, PWM, Sync, Formant Saw, WaveEdit Hienharm, WaveEdit Junox_ho, WaveEdit Euclidean, WaveEdit Sohler49, WaveEdit Vocal_fo, WaveEdit Pwn_saw, WaveEdit Tidyb030, WaveEdit I_heart, WaveEdit Sine_n, WaveEdit Vocal_00, WaveEdit Supersaw, WaveEdit Tidyb060, Consonant 024, WaveEdit Synlp92, WaveEdit Distorte, WaveEdit Fourier2, WaveEdit Grav_b8, WaveEdit Spectr01, WaveEdit Discorda, WaveEdit Sohler47, WaveEdit Ppg_wa27, WaveEdit Micro_q, WaveEdit Noise_wa, WaveEdit Kuato, WaveEdit Synlp67, WaveEdit Komple01, Consonant 066, WaveEdit Iterat00, WaveEdit Retro_sp, WaveEdit Acid_rin, WaveEdit Meravigl, WaveEdit Tidyb066, WaveEdit Tidyb049, WaveEdit Fracta03, WaveEdit Chebyshe, Sampled 167, WaveEdit Wavetabl, Chord 004, WaveEdit Grav_b6, WaveEdit Sohler33, WaveEdit Ppg_wa13, WaveEdit Pd104, WaveEdit Sohler48, WaveEdit Ppg_wa28, WaveEdit Wavetrip, WaveEdit Cz_ish, WaveEdit Fx_bitno, WaveEdit Tidyb046, WaveEdit Sohler74, WaveEdit Sohler29, WaveEdit Enshtu02, WaveEdit Tidyb072, WaveEdit Tidyb050, WaveEdit Voice_a, Resonator 018, WaveEdit Tidyb035, WaveEdit Tezzalog, AKWF 0004-hollow-01, WaveEdit Lom_a, WaveEdit Tidyb064, WaveEdit Moddrop, WaveEdit Harmomet, Hollow 013, WaveEdit Tidyb065, WaveEdit Ultra_ch, WaveEdit Laser_cr, WaveEdit Ppg_wa05, WaveEdit Drone, WaveEdit Gentle_m, WaveEdit Tidyb067, WaveEdit Tidyb005, Resonator 008, WaveEdit Synlp71, WaveEdit Sohler46, WaveEdit Tidyb077, WaveEdit Tidyb027, WaveEdit Sand_eye, WaveEdit Feedback, WaveEdit Tidyb071, WaveEdit Snake_ey, WaveEdit Hyperbol, AKWF 0018-hollow-01, Overtone 064, WaveEdit Light_00, WaveEdit Crush_ad, Organ 034, Sampled 131, Glass 012, WaveEdit	Vocal

Sohler67, Organ 006, WaveEdit Ppg_wa14, WaveEdit Sq8_sh, AKWF clavinet-hollow-01, Glass 009, WaveEdit Grav_c7, Consonant 100, WaveEdit Sohler51, Vowel Bass 015, Sampled 227, Hollow 005, Sampled 065, AKWF oscchip-04, AKWF raw-01, Hollow 015, AKWF snippets-01, Chord 027, Consonant 071, Chord 020, Consonant 051, WaveEdit Cyborg, Sampled 238, AKWF 0014-hollow-01, Overtone 000, WaveEdit Ppg_wa15, WaveEdit Grav_b4, Consonant 009, WaveEdit Grav_c5, WaveEdit Twist6, Glass 027, Sampled 078, AKWF oscchip-05, Consonant 120, WaveEdit Ppg_wa04, WaveEdit Sohler59, Sampled 081, Otmorph 053, Chord 030, Chord 032, Organ 029, Overtone 108, Hollow 001, WaveEdit Ppg_wa10, WaveEdit Organ_di, AKWF 0009-hollow-01, Tube 042, Vowel Bass 008, Organ 035, Metal 001, Consonant 114, AKWF vgame-03, Sampled 224, Resonator 001, Vowel Bass 038, Sampled 058, Vowel Tenor 038, Glass 026, Chord 019, Otmorph 115, AKWF stereo-04, Consonant 105, Vowel Bass 026, Chord 017, WaveEdit Ppg_wa11, Consonant 109, Sampled 125, Consonant 012, Overtone 046, Glass 016, Otmorph 067, Organ 026, Otmorph 005, Glass 018, Consonant 018, Hollow 023, Overtone 019, Otmorph 070, Consonant 127, Vowel Bass 029, Organ 016, Vowel Bass 017, Sampled 209, Sampled 130, Overtone 038, WaveEdit Reso_ana, Consonant 142, WaveEdit Reso_p00, Chord 009, AKWF distorted-01, Sampled 162, Glass 006, Metal 003, WaveEdit Square_p, Sampled 059, Organ 030, Sampled 003, Consonant 036, Chord 034, Sampled 193, WaveEdit Qux_fmy, Vowel Bass 023, Otmorph 133, WaveEdit Sohler52, Vowel Bass 007, WaveEdit Grav_b3, Consonant 025, Consonant 146, Chord 018, Overtone 047, Chord 038, Sampled 127, Metal 009, Organ 014, Consonant 033, Vowel Bass 035, Consonant 001, WaveEdit Iterativ, Consonant 129, Tube 066, Organ 025, Vowel Tenor 012, Chord 039, Sampled 106, Sampled 207, Breath 013, WaveEdit Grav_c2, Consonant 079, Otmorph 119, Consonant 064, WaveEdit Ppg_wa18, WaveEdit Pd103, Overtone 033, Chord 013, Vowel Bass 014, Sampled 226, Chord 011, Sampled 048, Vowel Tenor 037, Otmorph 095, Organ 012, Vowel Tenor 019, Overtone 015, Overtone 054, Consonant 088, Consonant 108, Sampled 197, Organ 037, Choir 004, Vowel Bass 030, Resonator 019, Otmorph 086, Tube 074, Tube 089, Sampled 182, Organ 038, Sampled 173, Consonant 060, Otmorph 030, Sampled 138, Consonant 056, Otmorph 132, Otmorph 138, Glass 003, Vowel Bass 021, Vowel Tenor 036, Organ 033, Sampled 214, Consonant 081, Consonant 094, Consonant 028, Consonant 003, Vowel Soprano 035, Consonant 126, Vowel Bass 027, Chord 000, Resonator 010, Choir 026, Metal 015, AKWF 0018-02, Consonant 046, Otmorph 079, Vowel Tenor 024, Consonant 117, Choir 025, Overtone 067, Chord 016, Otmorph 109, Vowel Alto 001, Otmorph 089, Consonant 077, Shimmer 033, Otmorph 022, Consonant 144, Organ 000, Consonant 139, Consonant 106, Vowel Bass 039, Otmorph 043, Vowel Tenor 033, Resonator 012, Otmorph 059, Otmorph 073, Sampled 093, Overtone 055, Sampled 189, Otmorph 068, Otmorph 064, Breath 029, Chord 033, Sampled 108, Consonant 030, Bowed 075, Organ 011, Vowel Tenor 006, Otmorph 056, AKWF 0006-02, Glass 037, Vowel Bass 001, WaveEdit Tidyb026, WaveEdit Sohler39, Sampled 076, Organ 021, WaveEdit Harmonio, WaveEdit Grav_c9, Consonant 037, WaveEdit Sine_a02, WaveEdit High_fre, WaveEdit Tidyb024, AKWF hvoice-01, WaveEdit Sohler58, WaveEdit Ppg_wa29, Vowel Bass 036, WaveEdit Sohler19, WaveEdit Sohler53, WaveEdit Sohler94, Glass 023, Chord 031, WaveEdit Access_v, Bowed 079, AKWF c604-01, WaveEdit Sohler56, WaveEdit Tidyb017, WaveEdit Sohler11, Overtone 018, Consonant 032, Reed 034, WaveEdit Mk_dwg_h, WaveEdit Grav_c3, WaveEdit Fracta01, WaveEdit Synlp54, WaveEdit Tidyb062, WaveEdit Envelo01, Vowel Tenor 005, WaveEdit Tidyb004, Chord 008, AKWF bitreduced-01, AKWF bw-saw-01, Overtone 053, WaveEdit Grav_a6, WaveEdit Sohler26, Sampled 066, WaveEdit Sohler86, Tube 036, Otmorph 014, WaveEdit Sohler76, WaveEdit Sohler27, Choir 030, WaveEdit

Bank_410, WaveEdit Tidyb018, WaveEdit Alpha_2, WaveEdit Tidyb003, WaveEdit Sohler35, WaveEdit Lfo_play, WaveEdit Ppg_wa01, WaveEdit Ppg_wa02, WaveEdit Prophe00, Vowel Tenor 011, Tube 072, WaveEdit Tidyb058, Consonant 137, WaveEdit Sohler31, Consonant 076, WaveEdit Just_ran, Consonant 080, WaveEdit Grav_a5, WaveEdit Sohler14, Resonator 020, WaveEdit Grav_a1, Glass 020, WaveEdit Sohler97, Chord 028, WaveEdit Grav_a2, Vowel Bass 034, WaveEdit Sohler70, Breath 031, AKWF oscchip-03, WaveEdit Sohler84, Consonant 149, WaveEdit Tidyb041, Consonant 023, Otmorph 126, Bowed 054, WaveEdit Mixed_as, Otmorph 084, WaveEdit Twist31, WaveEdit Spectr05, WaveEdit Sohler92, WaveEdit Isobelle, Chord 021, Consonant 067, WaveEdit Envelope, Sub 003, WaveEdit Sohler79, Tube 002, Consonant 008, WaveEdit Ppg_wa03, Pluck 028, Vowel Alto 012, AKWF epiano-01, AKWF bw-squ-02, Organ 005, WaveEdit Sine2saw, Vowel Alto 039, Glass 004, WaveEdit Sohler43, Sampled 109, Vowel Alto 003, Glass 010, AKWF clarinett-01, WaveEdit Sohler18, Sampled 170, Tube 046, Vowel Tenor 017, WaveEdit Foldfeed, Tube 008, AKWF 0015-01, Otmorph 106, Overtone 037, Consonant 022, Breath 019, WaveEdit Ppg_wa07, Sampled 164, WaveEdit Sine_mut, Vowel Alto 007, Sampled 211, Metal 025, Glass 007, Vowel Soprano 011, Glass 024, AKWF 0008-01, Vowel Soprano 033, Shimmer 002, Pure 015, WaveEdit Geometri, Sampled 203, Chord 007, Vowel Soprano 005, AKWF altosax-01, Vowel Soprano 015, WaveEdit Ppg_wa17, Overtone 010, WaveEdit Synlp159, Sampled 115, WaveEdit Ppg_wa22, Otmorph 112, Vowel Alto 016, Breath 033, Tube 009, Tube 079, Sampled 097, Chord 023, Stack 008, AKWF 0003-01, Glass 038, Metal 016

Position	lead.position	0 .. 1	0.3
Position Env	lead.pos_env	-1 .. 1	0
Position Decay	lead.pos_decay	10 .. 8000 ms	1500 ms
Position LFO	lead.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	lead.pos_lfo_beats	0.25 .. 64 beats	16 beats
Cutoff	lead.cutoff	200 .. 18000 Hz	9000 Hz
Resonance	lead.resonance	0 .. 1	0.1
Env Amount	lead.env_amount	0 .. 6 oct	1 oct
Filter Decay	lead.filter_decay	5 .. 3000 ms	600 ms
Key Track	lead.key_track	0 .. 1	0.5
HP Floor	lead.hp_floor	40 .. 400 Hz	250 Hz
HP Track	lead.hp_track	0 .. 1 x f0	1 x f0
Attack	lead.amp_attack	0.3 .. 2000 ms	3 ms
Decay	lead.amp_decay	5 .. 4000 ms	800 ms
Sustain	lead.amp_sustain	0 .. 1	0.85
Release	lead.amp_release	5 .. 4000 ms	220 ms
Width	lead.width	0 .. 1	0.8
Velocity	lead.vel_sens	0 .. 1	0.4

Delay Send	lead.delay_send	0 .. 1	0.22
Delay Left	lead.delay_left	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	1/4
Delay Right	lead.delay_right	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/8
Delay Feedback	lead.delay_feedback	0 .. 0.9	0.35
Delay High Pass	lead.delay_high_pass	150 .. 2000 Hz	350 Hz
Delay Low Pass	lead.delay_low_pass	800 .. 16000 Hz	6000 Hz
Room Send	lead.room_send	0 .. 1	0
Hall Send	lead.hall_send	0 .. 1	0.35
Duck	lead.duck	0 .. 18 dB	1.5 dB
Trance Gate	lead.gate	off / on	off
Gate Pattern	lead.gate_pattern	Sixteenths, Eighths, Rolling, Gallop, 3-3-2, Triplets, Trance 1, Trance 2, Trance 3	Trance 1
Gate Depth	lead.gate_depth	0 .. 1	0.85
Gate Duty	lead.gate_duty	0.05 .. 1	0.5
Gate Attack	lead.gate_attack	0.5 .. 60 ms	3 ms
Gate Release	lead.gate_release	0.5 .. 120 ms	12 ms
Gate Tone	lead.gate_tone	0 .. 1	0.4
Level	lead.level	-36 .. 6 dB	-9 dB
Disperse	lead.disperse	0 .. 8 x	0 x
Disperse Freq	lead.disperse_freq	200 .. 8000 Hz	1250 Hz
Drift	lead.drift	0 .. 8 ct	1 ct
Filter Type	lead.filter_type	Low Pass, Band Pass, High Pass, Notch	Low Pass
Glide	lead.glide	0 .. 400 ms	0 ms
Pan	lead.pan	-1 .. 1	-0.05
Oscillator 2	lead.osc2	Off, Supersaw, VA, FM, Wavetable	Supersaw
Osc 2 Mix	lead.osc2_mix	0 .. 1	0.35
Osc 2 Interval	lead.osc2_interval	-2 Oct, -1 Oct, -5th, Unison, +5th, +1 Oct	-1 Oct
Osc 2 Detune	lead.osc2_detune	-50 .. 50 ct	0 ct
Voice LFO	lead.lfo_beats	0.25 .. 64 beats	8 beats

LFO to Cutoff	lead.lfo_cutoff	0 .. 4 oct	0 oct
LFO to Pitch	lead.lfo_pitch	0 .. 50 ct	0 ct
LFO to Level	lead.lfo_amp	0 .. 1	0
Plate Send	lead.plate_send	0 .. 1	0.1
Slow Movement	lead.slow_mod	0 .. 1	0
Filter Model	lead.filter_model	State Variable, Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder, Korg35 (MS-20), Polivoks, Wasp	State Variable
Filter Mode	lead.filter_mode	0 .. 1	0
Filter Attack	lead.filt_attack	0.1 .. 4000 ms	0.1 ms
Filter Sustain	lead.filt_sustain	0 .. 1	0
Filter Release	lead.filt_release	5 .. 8000 ms	300 ms
Mod Attack	lead.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	lead.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	lead.menv_sustain	0 .. 1	0
Mod Release	lead.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	lead.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	lead.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	lead.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	lead.lfo1_retrig	off / on	off
LFO 1 Fade	lead.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	lead.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	lead.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	lead.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	lead.lfo2_retrig	off / on	off
LFO 2 Fade	lead.lfo2_fade	0 .. 8 s	0 s

LFO 3 Rate	lead.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	lead.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	lead.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrig	lead.lfo3_retrig	off / on	off
LFO 3 Fade	lead.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	lead.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	lead.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	lead.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrig	lead.lfo4_retrig	off / on	off
LFO 4 Fade	lead.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	lead.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	lead.mx1_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 1 Amount	lead.mx1_amount	-1 .. 1	0
Mod 2 Source	lead.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	lead.mx2_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 2 Amount	lead.mx2_amount	-1 .. 1	0
Mod 3 Source	lead.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	lead.mx3_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 3 Amount	lead.mx3_amount	-1 .. 1	0
Mod 4 Source	lead.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	lead.mx4_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 4 Amount	lead.mx4_amount	-1 .. 1	0
Mod 5 Source	lead.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off

Mod 5 Target	lead.mx5_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 5 Amount	lead.mx5_amount	-1 .. 1	0
Mod 6 Source	lead.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	lead.mx6_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 6 Amount	lead.mx6_amount	-1 .. 1	0
Mod 7 Source	lead.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	lead.mx7_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 7 Amount	lead.mx7_amount	-1 .. 1	0
Mod 8 Source	lead.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	lead.mx8_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 8 Amount	lead.mx8_amount	-1 .. 1	0

COUNTER

Control	Key	Range	Default
Oscillator	counter.osc	Supersaw, VA, FM, Wavetable	Supersaw
Detune	counter.detune	0 .. 1	0.45
Mix	counter.mix	0 .. 1	0.7
Dynamic Detune	counter.dynamic_detune	0 .. 1	0.6
Wave	counter.wave	0 .. 1	0
Pulse Width	counter.pulse_width	0.05 .. 0.95	0.5
FM Ratio	counter.fm_ratio	0.5 .. 8	2
FM Index	counter.fm_index	0 .. 10	2.5
FM Decay	counter.fm_decay	5 .. 2000 ms	250 ms
Table	counter.table	Classic, Vocal, Glass, PWM, Sync, Formant Saw, WaveEdit Hienharm, WaveEdit Junox_ho, WaveEdit Euclidean, WaveEdit Sohler49, WaveEdit Vocal_fo, WaveEdit Pwn_saw, WaveEdit Tidyb030, WaveEdit I_heart, WaveEdit Sine_n, WaveEdit Vocal_00, WaveEdit Supersaw, WaveEdit Tidyb060, Consonant 024, WaveEdit Synlp92, WaveEdit Distorte, WaveEdit Fourier2, WaveEdit Grav_b8, WaveEdit Spectr01, WaveEdit Discorda, WaveEdit Sohler47, WaveEdit Ppg_wa27, WaveEdit Micro_q, WaveEdit Noise_wa, WaveEdit Kuato, WaveEdit Synlp67, WaveEdit Komple01, Consonant 066, WaveEdit Iterat00, WaveEdit Retro_sp, WaveEdit Acid_rin, WaveEdit Meravigl, WaveEdit Tidyb066, WaveEdit Tidyb049, WaveEdit	Vocal

Fracta03, WaveEdit Chebyshe, Sampled 167, WaveEdit Wavetabl, Chord 004, WaveEdit Grav_b6, WaveEdit Sohler33, WaveEdit Ppg_wa13, WaveEdit Pd104, WaveEdit Sohler48, WaveEdit Ppg_wa28, WaveEdit Wavetrip, WaveEdit Cz_ish, WaveEdit Fx_bitno, WaveEdit Tidyb046, WaveEdit Sohler74, WaveEdit Sohler29, WaveEdit Enshtu02, WaveEdit Tidyb072, WaveEdit Tidyb050, WaveEdit Voice_a, Resonator 018, WaveEdit Tidyb035, WaveEdit Tezzalog, AKWF 0004-hollow-01, WaveEdit Lom_a, WaveEdit Tidyb064, WaveEdit Moddrop, WaveEdit Harmomet, Hollow 013, WaveEdit Tidyb065, WaveEdit Ultra_ch, WaveEdit Laser_cr, WaveEdit Ppg_wa05, WaveEdit Drone, WaveEdit Gentle_m, WaveEdit Tidyb067, WaveEdit Tidyb005, Resonator 008, WaveEdit Synlp71, WaveEdit Sohler46, WaveEdit Tidyb077, WaveEdit Tidyb027, WaveEdit Sand_eye, WaveEdit Feedback, WaveEdit Tidyb071, WaveEdit Snake_ey, WaveEdit Hyperbol, AKWF 0018-hollow-01, Overtone 064, WaveEdit Light_00, WaveEdit Crush_ad, Organ 034, Sampled 131, Glass 012, WaveEdit Sohler67, Organ 006, WaveEdit Ppg_wa14, WaveEdit Sq8_sh, AKWF clavinet-hollow-01, Glass 009, WaveEdit Grav_c7, Consonant 100, WaveEdit Sohler51, Vowel Bass 015, Sampled 227, Hollow 005, Sampled 065, AKWF oscchip-04, AKWF raw-01, Hollow 015, AKWF snippets-01, Chord 027, Consonant 071, Chord 020, Consonant 051, WaveEdit Cyborg, Sampled 238, AKWF 0014-hollow-01, Overtone 000, WaveEdit Ppg_wa15, WaveEdit Grav_b4, Consonant 009, WaveEdit Grav_c5, WaveEdit Twist6, Glass 027, Sampled 078, AKWF oscchip-05, Consonant 120, WaveEdit Ppg_wa04, WaveEdit Sohler59, Sampled 081, Otmorph 053, Chord 030, Chord 032, Organ 029, Overtone 108, Hollow 001, WaveEdit Ppg_wa10, WaveEdit Organ_di, AKWF 0009-hollow-01, Tube 042, Vowel Bass 008, Organ 035, Metal 001, Consonant 114, AKWF vgame-03, Sampled 224, Resonator 001, Vowel Bass 038, Sampled 058, Vowel Tenor 038, Glass 026, Chord 019, Otmorph 115, AKWF stereo-04, Consonant 105, Vowel Bass 026, Chord 017, WaveEdit Ppg_wa11, Consonant 109, Sampled 125, Consonant 012, Overtone 046, Glass 016, Otmorph 067, Organ 026, Otmorph 005, Glass 018, Consonant 018, Hollow 023, Overtone 019, Otmorph 070, Consonant 127, Vowel Bass 029, Organ 016, Vowel Bass 017, Sampled 209, Sampled 130, Overtone 038, WaveEdit Reso_ana, Consonant 142, WaveEdit Reso_p00, Chord 009, AKWF distorted-01, Sampled 162, Glass 006, Metal 003, WaveEdit Square_p, Sampled 059, Organ 030, Sampled 003, Consonant 036, Chord 034, Sampled 193, WaveEdit Qux_fmy, Vowel Bass 023, Otmorph 133, WaveEdit Sohler52, Vowel Bass 007, WaveEdit Grav_b3, Consonant 025, Consonant 146, Chord 018, Overtone 047, Chord 038, Sampled 127, Metal 009, Organ 014, Consonant 033, Vowel Bass 035, Consonant 001, WaveEdit Iterativ, Consonant 129, Tube 066, Organ 025, Vowel Tenor 012, Chord 039, Sampled 106, Sampled 207, Breath 013, WaveEdit Grav_c2, Consonant 079, Otmorph 119, Consonant 064, WaveEdit Ppg_wa18, WaveEdit Pd103, Overtone 033, Chord 013, Vowel Bass 014, Sampled 226, Chord 011, Sampled 048, Vowel Tenor 037, Otmorph 095, Organ 012, Vowel Tenor 019, Overtone 015, Overtone 054, Consonant 088, Consonant 108, Sampled 197, Organ 037, Choir 004, Vowel Bass 030, Resonator 019, Otmorph 086, Tube 074, Tube 089, Sampled 182, Organ 038, Sampled 173, Consonant 060, Otmorph 030, Sampled 138, Consonant 056, Otmorph 132, Otmorph 138, Glass 003, Vowel Bass 021, Vowel Tenor 036, Organ 033, Sampled 214, Consonant 081, Consonant 094, Consonant 028, Consonant 003, Vowel Soprano 035, Consonant 126, Vowel Bass 027, Chord 000, Resonator 010, Choir 026, Metal 015, AKWF 0018-02, Consonant 046, Otmorph 079, Vowel Tenor 024, Consonant 117, Choir 025, Overtone 067, Chord 016, Otmorph 109, Vowel Alto 001, Otmorph 089, Consonant 077, Shimmer 033, Otmorph 022, Consonant 144, Organ 000, Consonant

139, Consonant 106, Vowel Bass 039, Otmorph 043, Vowel Tenor 033, Resonator 012, Otmorph 059, Otmorph 073, Sampled 093, Overtone 055, Sampled 189, Otmorph 068, Otmorph 064, Breath 029, Chord 033, Sampled 108, Consonant 030, Bowed 075, Organ 011, Vowel Tenor 006, Otmorph 056, AKWF 0006-02, Glass 037, Vowel Bass 001, WaveEdit Tidyb026, WaveEdit Sohler39, Sampled 076, Organ 021, WaveEdit Harmonio, WaveEdit Grav_c9, Consonant 037, WaveEdit Sine_a02, WaveEdit High_fre, WaveEdit Tidyb024, AKWF hvoice-01, WaveEdit Sohler58, WaveEdit Ppg_wa29, Vowel Bass 036, WaveEdit Sohler19, WaveEdit Sohler53, WaveEdit Sohler94, Glass 023, Chord 031, WaveEdit Access_v, Bowed 079, AKWF c604-01, WaveEdit Sohler56, WaveEdit Tidyb017, WaveEdit Sohler11, Overtone 018, Consonant 032, Reed 034, WaveEdit Mk_dwg_h, WaveEdit Grav_c3, WaveEdit Fracta01, WaveEdit Synlp54, WaveEdit Tidyb062, WaveEdit Envelo01, Vowel Tenor 005, WaveEdit Tidyb004, Chord 008, AKWF bitreduced-01, AKWF bw-saw-01, Overtone 053, WaveEdit Grav_a6, WaveEdit Sohler26, Sampled 066, WaveEdit Sohler86, Tube 036, Otmorph 014, WaveEdit Sohler76, WaveEdit Sohler27, Choir 030, WaveEdit Bank_410, WaveEdit Tidyb018, WaveEdit Alpha_2, WaveEdit Tidyb003, WaveEdit Sohler35, WaveEdit Lfo_play, WaveEdit Ppg_wa01, WaveEdit Ppg_wa02, WaveEdit Prophe00, Vowel Tenor 011, Tube 072, WaveEdit Tidyb058, Consonant 137, WaveEdit Sohler31, Consonant 076, WaveEdit Just_ran, Consonant 080, WaveEdit Grav_a5, WaveEdit Sohler14, Resonator 020, WaveEdit Grav_a1, Glass 020, WaveEdit Sohler97, Chord 028, WaveEdit Grav_a2, Vowel Bass 034, WaveEdit Sohler70, Breath 031, AKWF oscchip-03, WaveEdit Sohler84, Consonant 149, WaveEdit Tidyb041, Consonant 023, Otmorph 126, Bowed 054, WaveEdit Mixed_as, Otmorph 084, WaveEdit Twist31, WaveEdit Spectr05, WaveEdit Sohler92, WaveEdit Isobelle, Chord 021, Consonant 067, WaveEdit Envelope, Sub 003, WaveEdit Sohler79, Tube 002, Consonant 008, WaveEdit Ppg_wa03, Pluck 028, Vowel Alto 012, AKWF epiano-01, AKWF bw-squ-02, Organ 005, WaveEdit Sine2saw, Vowel Alto 039, Glass 004, WaveEdit Sohler43, Sampled 109, Vowel Alto 003, Glass 010, AKWF clarinett-01, WaveEdit Sohler18, Sampled 170, Tube 046, Vowel Tenor 017, WaveEdit Foldfeed, Tube 008, AKWF 0015-01, Otmorph 106, Overtone 037, Consonant 022, Breath 019, WaveEdit Ppg_wa07, Sampled 164, WaveEdit Sine_mut, Vowel Alto 007, Sampled 211, Metal 025, Glass 007, Vowel Soprano 011, Glass 024, AKWF 0008-01, Vowel Soprano 033, Shimmer 002, Pure 015, WaveEdit Geometri, Sampled 203, Chord 007, Vowel Soprano 005, AKWF altosax-01, Vowel Soprano 015, WaveEdit Ppg_wa17, Overtone 010, WaveEdit Synlp159, Sampled 115, WaveEdit Ppg_wa22, Otmorph 112, Vowel Alto 016, Breath 033, Tube 009, Tube 079, Sampled 097, Chord 023, Stack 008, AKWF 0003-01, Glass 038, Metal 016

Position	counter.position	0 .. 1	0.3
Position Env	counter.pos_env	-1 .. 1	0
Position Decay	counter.pos_decay	10 .. 8000 ms	1500 ms
Position LFO	counter.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	counter.pos_lfo_beats	0.25 .. 64 beats	16 beats
Cutoff	counter.cutoff	200 .. 18000 Hz	7000 Hz
Resonance	counter.resonance	0 .. 1	0.15

Env Amount	counter.env_amount	0 .. 6 oct	2 oct
Filter Decay	counter.filter_decay	5 .. 3000 ms	400 ms
Key Track	counter.key_track	0 .. 1	0.5
HP Floor	counter.hp_floor	40 .. 400 Hz	400 Hz
HP Track	counter.hp_track	0 .. 1 x f0	1 x f0
Attack	counter.amp_attack	0.3 .. 2000 ms	5 ms
Decay	counter.amp_decay	5 .. 4000 ms	500 ms
Sustain	counter.amp_sustain	0 .. 1	0.8
Release	counter.amp_release	5 .. 4000 ms	180 ms
Width	counter.width	0 .. 1	0.8
Velocity	counter.vel_sens	0 .. 1	0.4
Delay Send	counter.delay_send	0 .. 1	0.15
Delay Left	counter.delay_left	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Delay Right	counter.delay_right	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	1/4
Delay Feedback	counter.delay_feedback	0 .. 0.9	0.4
Delay High Pass	counter.delay_high_pass	150 .. 2000 Hz	350 Hz
Delay Low Pass	counter.delay_low_pass	800 .. 16000 Hz	6000 Hz
Room Send	counter.room_send	0 .. 1	0
Hall Send	counter.hall_send	0 .. 1	0.25
Duck	counter.duck	0 .. 18 dB	1.5 dB
Trance Gate	counter.gate	off / on	off
Gate Pattern	counter.gate_pattern	Sixteenths, Eighths, Rolling, Gallop, 3-3-2, Triplets, Trance 1, Trance 2, Trance 3	Trance 1
Gate Depth	counter.gate_depth	0 .. 1	0.85
Gate Duty	counter.gate_duty	0.05 .. 1	0.5
Gate Attack	counter.gate_attack	0.5 .. 60 ms	3 ms
Gate Release	counter.gate_release	0.5 .. 120 ms	12 ms
Gate Tone	counter.gate_tone	0 .. 1	0.4
Level	counter.level	-36 .. 6 dB	-12 dB
Disperse	counter.disperse	0 .. 8 x	0 x

Disperse Freq	counter.disperse_freq	200 .. 8000 Hz	1250 Hz
Drift	counter.drift	0 .. 8 ct	1 ct
Filter Type	counter.filter_type	Low Pass, Band Pass, High Pass, Notch	Low Pass
Glide	counter.glide	0 .. 400 ms	0 ms
Pan	counter.pan	-1 .. 1	0.1
Oscillator 2	counter.osc2	Off, Supersaw, VA, FM, Wavetable	Off
Osc 2 Mix	counter.osc2_mix	0 .. 1	0
Osc 2 Interval	counter.osc2_interval	-2 Oct, -1 Oct, -5th, Unison, +5th, +1 Oct	-1 Oct
Osc 2 Detune	counter.osc2_detune	-50 .. 50 ct	0 ct
Voice LFO	counter.lfo_beats	0.25 .. 64 beats	8 beats
LFO to Cutoff	counter.lfo_cutoff	0 .. 4 oct	0 oct
LFO to Pitch	counter.lfo_pitch	0 .. 50 ct	0 ct
LFO to Level	counter.lfo_amp	0 .. 1	0
Plate Send	counter.plate_send	0 .. 1	0
Slow Movement	counter.slow_mod	0 .. 1	0
Filter Model	counter.filter_model	State Variable, Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder, Korg35 (MS-20), Polivoks, Wasp	State Variable
Filter Mode	counter.filter_mode	0 .. 1	0
Filter Attack	counter.filt_attack	0.1 .. 4000 ms	0.1 ms
Filter Sustain	counter.filt_sustain	0 .. 1	0
Filter Release	counter.filt_release	5 .. 8000 ms	300 ms
Mod Attack	counter.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	counter.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	counter.menv_sustain	0 .. 1	0
Mod Release	counter.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	counter.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	counter.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine

LFO 1 Sync	counter.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	counter.lfo1_retrig	off / on	off
LFO 1 Fade	counter.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	counter.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	counter.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	counter.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	counter.lfo2_retrig	off / on	off
LFO 2 Fade	counter.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	counter.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	counter.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	counter.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	counter.lfo3_retrig	off / on	off
LFO 3 Fade	counter.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	counter.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	counter.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	counter.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	counter.lfo4_retrig	off / on	off
LFO 4 Fade	counter.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	counter.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	counter.mx1_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 1 Amount	counter.mx1_amount	-1 .. 1	0
Mod 2 Source	counter.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	counter.mx2_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 2 Amount	counter.mx2_amount	-1 .. 1	0

Mod 3 Source	counter.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	counter.mx3_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 3 Amount	counter.mx3_amount	-1 .. 1	0
Mod 4 Source	counter.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	counter.mx4_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 4 Amount	counter.mx4_amount	-1 .. 1	0
Mod 5 Source	counter.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	counter.mx5_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 5 Amount	counter.mx5_amount	-1 .. 1	0
Mod 6 Source	counter.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	counter.mx6_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 6 Amount	counter.mx6_amount	-1 .. 1	0
Mod 7 Source	counter.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	counter.mx7_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 7 Amount	counter.mx7_amount	-1 .. 1	0
Mod 8 Source	counter.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	counter.mx8_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 8 Amount	counter.mx8_amount	-1 .. 1	0

PLUCK

Control	Key	Range	Default
Oscillator	pluck.osc	Supersaw, VA, FM, Wavetable	VA
Detune	pluck.detune	0 .. 1	0.3
Mix	pluck.mix	0 .. 1	0.55
Dynamic Detune	pluck.dynamic_detune	0 .. 1	0.6
Wave	pluck.wave	0 .. 1	0.15

Pulse Width	pluck.pulse_width	0.05 .. 0.95	0.5
FM Ratio	pluck.fm_ratio	0.5 .. 8	2
FM Index	pluck.fm_index	0 .. 10	2.5
FM Decay	pluck.fm_decay	5 .. 2000 ms	250 ms
Table	pluck.table	Classic, Vocal, Glass, PWM, Sync, Formant Saw, WaveEdit Hienharm, WaveEdit Junox_ho, WaveEdit Euclidean, WaveEdit Sohler49, WaveEdit Vocal_fo, WaveEdit Pwn_saw, WaveEdit Tidyb030, WaveEdit I_heart, WaveEdit Sine_n, WaveEdit Vocal_00, WaveEdit Supersaw, WaveEdit Tidyb060, Consonant 024, WaveEdit Synlp92, WaveEdit Distorte, WaveEdit Fourier2, WaveEdit Grav_b8, WaveEdit Spectr01, WaveEdit Discorda, WaveEdit Sohler47, WaveEdit Ppg_wa27, WaveEdit Micro_q, WaveEdit Noise_wa, WaveEdit Kuato, WaveEdit Synlp67, WaveEdit Komple01, Consonant 066, WaveEdit Iterat00, WaveEdit Retro_sp, WaveEdit Acid_rin, WaveEdit Meravigl, WaveEdit Tidyb066, WaveEdit Tidyb049, WaveEdit Fracta03, WaveEdit Chebyshe, Sampled 167, WaveEdit Wavetabl, Chord 004, WaveEdit Grav_b6, WaveEdit Sohler33, WaveEdit Ppg_wa13, WaveEdit Pd104, WaveEdit Sohler48, WaveEdit Ppg_wa28, WaveEdit Wavetrip, WaveEdit Cz_ish, WaveEdit Fx_bitno, WaveEdit Tidyb046, WaveEdit Sohler74, WaveEdit Sohler29, WaveEdit Enshtu02, WaveEdit Tidyb072, WaveEdit Tidyb050, WaveEdit Voice_a, Resonator 018, WaveEdit Tidyb035, WaveEdit Tezzalog, AKWF 0004-hollow-01, WaveEdit Lom_a, WaveEdit Tidyb064, WaveEdit Moddrop, WaveEdit Harmomet, Hollow 013, WaveEdit Tidyb065, WaveEdit Ultra_ch, WaveEdit Laser_cr, WaveEdit Ppg_wa05, WaveEdit Drone, WaveEdit Gentle_m, WaveEdit Tidyb067, WaveEdit Tidyb005, Resonator 008, WaveEdit Synlp71, WaveEdit Sohler46, WaveEdit Tidyb077, WaveEdit Tidyb027, WaveEdit Sand_eye, WaveEdit Feedback, WaveEdit Tidyb071, WaveEdit Snake_ey, WaveEdit Hyperbol, AKWF 0018-hollow-01, Overtone 064, WaveEdit Light_00, WaveEdit Crush_ad, Organ 034, Sampled 131, Glass 012, WaveEdit Sohler67, Organ 006, WaveEdit Ppg_wa14, WaveEdit Sq8_sh, AKWF clavinet-hollow-01, Glass 009, WaveEdit Grav_c7, Consonant 100, WaveEdit Sohler51, Vowel Bass 015, Sampled 227, Hollow 005, Sampled 065, AKWF oscchip-04, AKWF raw-01, Hollow 015, AKWF snippets-01, Chord 027, Consonant 071, Chord 020, Consonant 051, WaveEdit Cyborg, Sampled 238, AKWF 0014-hollow-01, Overtone 000, WaveEdit Ppg_wa15, WaveEdit Grav_b4, Consonant 009, WaveEdit Grav_c5, WaveEdit Twist6, Glass 027, Sampled 078, AKWF oscchip-05, Consonant 120, WaveEdit Ppg_wa04, WaveEdit Sohler59, Sampled 081, Otmorph 053, Chord 030, Chord 032, Organ 029, Overtone 108, Hollow 001, WaveEdit Ppg_wa10, WaveEdit Organ_di, AKWF 0009-hollow-01, Tube 042, Vowel Bass 008, Organ 035, Metal 001, Consonant 114, AKWF vgame-03, Sampled 224, Resonator 001, Vowel Bass 038, Sampled 058, Vowel Tenor 038, Glass 026, Chord 019, Otmorph 115, AKWF stereo-04, Consonant 105, Vowel Bass 026, Chord 017, WaveEdit Ppg_wa11, Consonant 109, Sampled 125, Consonant 012, Overtone 046, Glass 016, Otmorph 067, Organ 026, Otmorph 005, Glass 018, Consonant 018, Hollow 023, Overtone 019, Otmorph 070, Consonant 127, Vowel Bass 029, Organ 016, Vowel Bass 017, Sampled 209, Sampled 130, Overtone 038, WaveEdit Reso_ana, Consonant 142, WaveEdit Reso_p00, Chord 009, AKWF distorted-01, Sampled 162, Glass 006, Metal 003, WaveEdit Square_p, Sampled 059, Organ 030, Sampled 003, Consonant 036, Chord 034, Sampled 193, WaveEdit Qux_fmy, Vowel Bass 023, Otmorph 133, WaveEdit Sohler52, Vowel Bass 007, WaveEdit Grav_b3, Consonant 025, Consonant 146, Chord 018, Overtone 047, Chord 038, Sampled 127, Metal 009, Organ 014, Consonant 033, Vowel Bass 035, Consonant 001, WaveEdit Iterativ, Consonant 129, Tube 066, Organ 025, Vowel Tenor 012, Chord 039, Sampled	Vocal

106, Sampled 207, Breath 013, WaveEdit Grav_c2, Consonant 079, Otmorph 119, Consonant 064, WaveEdit Ppg_wa18, WaveEdit Pd103, Overtone 033, Chord 013, Vowel Bass 014, Sampled 226, Chord 011, Sampled 048, Vowel Tenor 037, Otmorph 095, Organ 012, Vowel Tenor 019, Overtone 015, Overtone 054, Consonant 088, Consonant 108, Sampled 197, Organ 037, Choir 004, Vowel Bass 030, Resonator 019, Otmorph 086, Tube 074, Tube 089, Sampled 182, Organ 038, Sampled 173, Consonant 060, Otmorph 030, Sampled 138, Consonant 056, Otmorph 132, Otmorph 138, Glass 003, Vowel Bass 021, Vowel Tenor 036, Organ 033, Sampled 214, Consonant 081, Consonant 094, Consonant 028, Consonant 003, Vowel Soprano 035, Consonant 126, Vowel Bass 027, Chord 000, Resonator 010, Choir 026, Metal 015, AKWF 0018-02, Consonant 046, Otmorph 079, Vowel Tenor 024, Consonant 117, Choir 025, Overtone 067, Chord 016, Otmorph 109, Vowel Alto 001, Otmorph 089, Consonant 077, Shimmer 033, Otmorph 022, Consonant 144, Organ 000, Consonant 139, Consonant 106, Vowel Bass 039, Otmorph 043, Vowel Tenor 033, Resonator 012, Otmorph 059, Otmorph 073, Sampled 093, Overtone 055, Sampled 189, Otmorph 068, Otmorph 064, Breath 029, Chord 033, Sampled 108, Consonant 030, Bowed 075, Organ 011, Vowel Tenor 006, Otmorph 056, AKWF 0006-02, Glass 037, Vowel Bass 001, WaveEdit Tidyb026, WaveEdit Sohler39, Sampled 076, Organ 021, WaveEdit Harmonio, WaveEdit Grav_c9, Consonant 037, WaveEdit Sine_a02, WaveEdit High_fre, WaveEdit Tidyb024, AKWF hvoice-01, WaveEdit Sohler58, WaveEdit Ppg_wa29, Vowel Bass 036, WaveEdit Sohler19, WaveEdit Sohler53, WaveEdit Sohler94, Glass 023, Chord 031, WaveEdit Access_v, Bowed 079, AKWF c604-01, WaveEdit Sohler56, WaveEdit Tidyb017, WaveEdit Sohler11, Overtone 018, Consonant 032, Reed 034, WaveEdit Mk_dwg_h, WaveEdit Grav_c3, WaveEdit Fracta01, WaveEdit Synlp54, WaveEdit Tidyb062, WaveEdit Envelo01, Vowel Tenor 005, WaveEdit Tidyb004, Chord 008, AKWF bitreduced-01, AKWF bw-saw-01, Overtone 053, WaveEdit Grav_a6, WaveEdit Sohler26, Sampled 066, WaveEdit Sohler86, Tube 036, Otmorph 014, WaveEdit Sohler76, WaveEdit Sohler27, Choir 030, WaveEdit Bank_410, WaveEdit Tidyb018, WaveEdit Alpha_2, WaveEdit Tidyb003, WaveEdit Sohler35, WaveEdit Lfo_play, WaveEdit Ppg_wa01, WaveEdit Ppg_wa02, WaveEdit Prophe00, Vowel Tenor 011, Tube 072, WaveEdit Tidyb058, Consonant 137, WaveEdit Sohler31, Consonant 076, WaveEdit Just_ran, Consonant 080, WaveEdit Grav_a5, WaveEdit Sohler14, Resonator 020, WaveEdit Grav_a1, Glass 020, WaveEdit Sohler97, Chord 028, WaveEdit Grav_a2, Vowel Bass 034, WaveEdit Sohler70, Breath 031, AKWF oscchip-03, WaveEdit Sohler84, Consonant 149, WaveEdit Tidyb041, Consonant 023, Otmorph 126, Bowed 054, WaveEdit Mixed_as, Otmorph 084, WaveEdit Twist31, WaveEdit Spectr05, WaveEdit Sohler92, WaveEdit Isobelle, Chord 021, Consonant 067, WaveEdit Envelope, Sub 003, WaveEdit Sohler79, Tube 002, Consonant 008, WaveEdit Ppg_wa03, Pluck 028, Vowel Alto 012, AKWF epiano-01, AKWF bw-squ-02, Organ 005, WaveEdit Sine2saw, Vowel Alto 039, Glass 004, WaveEdit Sohler43, Sampled 109, Vowel Alto 003, Glass 010, AKWF clarinett-01, WaveEdit Sohler18, Sampled 170, Tube 046, Vowel Tenor 017, WaveEdit Foldfeed, Tube 008, AKWF 0015-01, Otmorph 106, Overtone 037, Consonant 022, Breath 019, WaveEdit Ppg_wa07, Sampled 164, WaveEdit Sine_mut, Vowel Alto 007, Sampled 211, Metal 025, Glass 007, Vowel Soprano 011, Glass 024, AKWF 0008-01, Vowel Soprano 033, Shimmer 002, Pure 015, WaveEdit Geometri, Sampled 203, Chord 007, Vowel Soprano 005, AKWF altosax-01, Vowel Soprano 015, WaveEdit Ppg_wa17, Overtone 010, WaveEdit Synlp159, Sampled 115, WaveEdit Ppg_wa22, Otmorph 112, Vowel Alto 016, Breath 033, Tube 009, Tube 079, Sampled 097, Chord 023, Stack 008, AKWF 0003-01, Glass 038, Metal 016

Position Env	<code>pluck.pos_env</code>	-1 .. 1	0
Position Decay	<code>pluck.pos_decay</code>	10 .. 8000 ms	1500 ms
Position LFO	<code>pluck.pos_lfo_depth</code>	0 .. 0.5	0.15
LFO Period	<code>pluck.pos_lfo_beats</code>	0.25 .. 64 beats	16 beats
Cutoff	<code>pluck.cutoff</code>	200 .. 18000 Hz	1800 Hz
Resonance	<code>pluck.resonance</code>	0 .. 1	0.2
Env Amount	<code>pluck.env_amount</code>	0 .. 6 oct	3.2 oct
Filter Decay	<code>pluck.filter_decay</code>	5 .. 3000 ms	180 ms
Key Track	<code>pluck.key_track</code>	0 .. 1	0.5
HP Floor	<code>pluck.hp_floor</code>	40 .. 400 Hz	200 Hz
HP Track	<code>pluck.hp_track</code>	0 .. 1 x f0	1 x f0
Attack	<code>pluck.amp_attack</code>	0.3 .. 2000 ms	0.3 ms
Decay	<code>pluck.amp_decay</code>	5 .. 4000 ms	260 ms
Sustain	<code>pluck.amp_sustain</code>	0 .. 1	0
Release	<code>pluck.amp_release</code>	5 .. 4000 ms	200 ms
Width	<code>pluck.width</code>	0 .. 1	0.9
Velocity	<code>pluck.vel_sens</code>	0 .. 1	0.4
Delay Send	<code>pluck.delay_send</code>	0 .. 1	0.22
Delay Left	<code>pluck.delay_left</code>	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Delay Right	<code>pluck.delay_right</code>	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	1/8
Delay Feedback	<code>pluck.delay_feedback</code>	0 .. 0.9	0.25
Delay High Pass	<code>pluck.delay_high_pass</code>	150 .. 2000 Hz	350 Hz
Delay Low Pass	<code>pluck.delay_low_pass</code>	800 .. 16000 Hz	6000 Hz
Room Send	<code>pluck.room_send</code>	0 .. 1	0.25
Hall Send	<code>pluck.hall_send</code>	0 .. 1	0.05
Duck	<code>pluck.duck</code>	0 .. 18 dB	3 dB
Trance Gate	<code>pluck.gate</code>	off / on	off
Gate Pattern	<code>pluck.gate_pattern</code>	Sixteenths, Eighths, Rolling, Gallop, 3-3-2, Triplets, Trance 1, Trance 2, Trance 3	Trance 1

Gate Depth	pluck.gate_depth	0 .. 1	0.85
Gate Duty	pluck.gate_duty	0.05 .. 1	0.5
Gate Attack	pluck.gate_attack	0.5 .. 60 ms	3 ms
Gate Release	pluck.gate_release	0.5 .. 120 ms	12 ms
Gate Tone	pluck.gate_tone	0 .. 1	0.4
Level	pluck.level	-36 .. 6 dB	-8 dB
Disperse	pluck.disperse	0 .. 8 x	0 x
Disperse Freq	pluck.disperse_freq	200 .. 8000 Hz	1250 Hz
Drift	pluck.drift	0 .. 8 ct	1 ct
Filter Type	pluck.filter_type	Low Pass, Band Pass, High Pass, Notch	Low Pass
Glide	pluck.glide	0 .. 400 ms	0 ms
Pan	pluck.pan	-1 .. 1	0
Oscillator 2	pluck.osc2	Off, Supersaw, VA, FM, Wavetable	Off
Osc 2 Mix	pluck.osc2_mix	0 .. 1	0
Osc 2 Interval	pluck.osc2_interval	-2 Oct, -1 Oct, -5th, Unison, +5th, +1 Oct	-1 Oct
Osc 2 Detune	pluck.osc2_detune	-50 .. 50 ct	0 ct
Voice LFO	pluck.lfo_beats	0.25 .. 64 beats	8 beats
LFO to Cutoff	pluck.lfo_cutoff	0 .. 4 oct	0 oct
LFO to Pitch	pluck.lfo_pitch	0 .. 50 ct	0 ct
LFO to Level	pluck.lfo_amp	0 .. 1	0
Plate Send	pluck.plate_send	0 .. 1	0.05
Slow Movement	pluck.slow_mod	0 .. 1	0
Filter Model	pluck.filter_model	State Variable, Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder, Korg35 (MS-20), Polivoks, Wasp	State Variable
Filter Mode	pluck.filter_mode	0 .. 1	0
Filter Attack	pluck.filt_attack	0.1 .. 4000 ms	0.1 ms
Filter Sustain	pluck.filt_sustain	0 .. 1	0
Filter Release	pluck.filt_release	5 .. 8000 ms	300 ms

Mod Attack	<code>pluck.menv_attack</code>	0.1 .. 8000 ms	10 ms
Mod Decay	<code>pluck.menv_decay</code>	5 .. 12000 ms	800 ms
Mod Sustain	<code>pluck.menv_sustain</code>	0 .. 1	0
Mod Release	<code>pluck.menv_release</code>	5 .. 12000 ms	400 ms
LFO 1 Rate	<code>pluck.lfo1_rate</code>	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	<code>pluck.lfo1_shape</code>	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	<code>pluck.lfo1_sync</code>	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	<code>pluck.lfo1_retrigger</code>	off / on	off
LFO 1 Fade	<code>pluck.lfo1_fade</code>	0 .. 8 s	0 s
LFO 2 Rate	<code>pluck.lfo2_rate</code>	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	<code>pluck.lfo2_shape</code>	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	<code>pluck.lfo2_sync</code>	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	<code>pluck.lfo2_retrigger</code>	off / on	off
LFO 2 Fade	<code>pluck.lfo2_fade</code>	0 .. 8 s	0 s
LFO 3 Rate	<code>pluck.lfo3_rate</code>	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	<code>pluck.lfo3_shape</code>	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	<code>pluck.lfo3_sync</code>	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	<code>pluck.lfo3_retrigger</code>	off / on	off
LFO 3 Fade	<code>pluck.lfo3_fade</code>	0 .. 8 s	0 s
LFO 4 Rate	<code>pluck.lfo4_rate</code>	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	<code>pluck.lfo4_shape</code>	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	<code>pluck.lfo4_sync</code>	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	<code>pluck.lfo4_retrigger</code>	off / on	off
LFO 4 Fade	<code>pluck.lfo4_fade</code>	0 .. 8 s	0 s

Mod 1 Source	pluck.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	pluck.mx1_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 1 Amount	pluck.mx1_amount	-1 .. 1	0
Mod 2 Source	pluck.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	pluck.mx2_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 2 Amount	pluck.mx2_amount	-1 .. 1	0
Mod 3 Source	pluck.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	pluck.mx3_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 3 Amount	pluck.mx3_amount	-1 .. 1	0
Mod 4 Source	pluck.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	pluck.mx4_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 4 Amount	pluck.mx4_amount	-1 .. 1	0
Mod 5 Source	pluck.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	pluck.mx5_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 5 Amount	pluck.mx5_amount	-1 .. 1	0
Mod 6 Source	pluck.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	pluck.mx6_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 6 Amount	pluck.mx6_amount	-1 .. 1	0
Mod 7 Source	pluck.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	pluck.mx7_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 7 Amount	pluck.mx7_amount	-1 .. 1	0
Mod 8 Source	pluck.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	pluck.mx8_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off

Mod 8 Amount	pluck.mx8_amount	-1 .. 1	0
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ARP

Control	Key	Range	Default
Oscillator	arp.osc	Supersaw, VA, FM, Wavetable	VA
Detune	arp.detune	0 .. 1	0.2
Mix	arp.mix	0 .. 1	0.5
Dynamic Detune	arp.dynamic_detune	0 .. 1	0.6
Wave	arp.wave	0 .. 1	0.1
Pulse Width	arp.pulse_width	0.05 .. 0.95	0.5
FM Ratio	arp.fm_ratio	0.5 .. 8	2
FM Index	arp.fm_index	0 .. 10	2.5
FM Decay	arp.fm_decay	5 .. 2000 ms	250 ms
Table	arp.table	Classic, Vocal, Glass, PWM, Sync, Formant Saw, WaveEdit Hienharm, WaveEdit Junox_ho, WaveEdit Euclidea, WaveEdit Sohler49, WaveEdit Vocal_fo, WaveEdit Pwn_saw, WaveEdit Tidyb030, WaveEdit I_heart, WaveEdit Sine_n, WaveEdit Vocal_00, WaveEdit Supersaw, WaveEdit Tidyb060, Consonant 024, WaveEdit Synlp92, WaveEdit Distorte, WaveEdit Fourier2, WaveEdit Grav_b8, WaveEdit Spectr01, WaveEdit Discorda, WaveEdit Sohler47, WaveEdit Ppg_wa27, WaveEdit Micro_q, WaveEdit Noise_wa, WaveEdit Kuato, WaveEdit Synlp67, WaveEdit Komple01, Consonant 066, WaveEdit Iterat00, WaveEdit Retro_sp, WaveEdit Acid_rin, WaveEdit Meravigl, WaveEdit Tidyb066, WaveEdit Tidyb049, WaveEdit Fracta03, WaveEdit Chebyshe, Sampled 167, WaveEdit Wavetabl, Chord 004, WaveEdit Grav_b6, WaveEdit Sohler33, WaveEdit Ppg_wa13, WaveEdit Pd104, WaveEdit Sohler48, WaveEdit Ppg_wa28, WaveEdit Wavetrip, WaveEdit Cz_ish, WaveEdit Fx_bitno, WaveEdit Tidyb046, WaveEdit Sohler74, WaveEdit Sohler29, WaveEdit Enshtu02, WaveEdit Tidyb072, WaveEdit Tidyb050, WaveEdit Voice_a, Resonator 018, WaveEdit Tidyb035, WaveEdit Tezzalog, AKWF 0004-hollow-01, WaveEdit Lom_a, WaveEdit Tidyb064, WaveEdit Moddrop, WaveEdit Harmomet, Hollow 013, WaveEdit Tidyb065, WaveEdit Ultra_ch, WaveEdit Laser_cr, WaveEdit Ppg_wa05, WaveEdit Drone, WaveEdit Gentle_m, WaveEdit Tidyb067, WaveEdit Tidyb005, Resonator 008, WaveEdit Synlp71, WaveEdit Sohler46, WaveEdit Tidyb077, WaveEdit Tidyb027, WaveEdit Sand_eye, WaveEdit Feedback, WaveEdit Tidyb071, WaveEdit Snake_ey, WaveEdit Hyperbol, AKWF 0018-hollow-01, Overtone 064, WaveEdit Light_00, WaveEdit Crush_ad, Organ 034, Sampled 131, Glass 012, WaveEdit Sohler67, Organ 006, WaveEdit Ppg_wa14, WaveEdit Sq8_sh, AKWF clavinet-hollow-01, Glass 009, WaveEdit Grav_c7, Consonant 100, WaveEdit Sohler51, Vowel Bass 015, Sampled 227, Hollow 005, Sampled 065, AKWF oscchip-04, AKWF raw-01, Hollow 015, AKWF snippets-01, Chord 027, Consonant 071, Chord 020, Consonant 051, WaveEdit Cyborg, Sampled 238, AKWF 0014-hollow-01, Overtone 000, WaveEdit Ppg_wa15, WaveEdit Grav_b4, Consonant 009, WaveEdit Grav_c5, WaveEdit Twist6, Glass 027, Sampled 078, AKWF oscchip-05, Consonant 120, WaveEdit Ppg_wa04, WaveEdit Sohler59, Sampled 081, Otmorph 053, Chord 030, Chord 032, Organ 029, Overtone 108, Hollow 001, WaveEdit Ppg_wa10, WaveEdit Organ_di, AKWF 0009-hollow-01, Tube 042, Vowel Bass 008, Organ 035, Metal 001, Consonant 114, AKWF vgame-03, Sampled 224, Resonator 001, Vowel Bass 038, Sampled 058, Vowel Tenor 038, Glass 026, Chord 019,	Vocal

Otmorph 115, AKWF stereo-04, Consonant 105, Vowel Bass 026, Chord 017, WaveEdit Ppg_wa11, Consonant 109, Sampled 125, Consonant 012, Overtone 046, Glass 016, Otmorph 067, Organ 026, Otmorph 005, Glass 018, Consonant 018, Hollow 023, Overtone 019, Otmorph 070, Consonant 127, Vowel Bass 029, Organ 016, Vowel Bass 017, Sampled 209, Sampled 130, Overtone 038, WaveEdit Reso_ana, Consonant 142, WaveEdit Reso_p00, Chord 009, AKWF distorted-01, Sampled 162, Glass 006, Metal 003, WaveEdit Square_p, Sampled 059, Organ 030, Sampled 003, Consonant 036, Chord 034, Sampled 193, WaveEdit Qux_fmy, Vowel Bass 023, Otmorph 133, WaveEdit Sohler52, Vowel Bass 007, WaveEdit Grav_b3, Consonant 025, Consonant 146, Chord 018, Overtone 047, Chord 038, Sampled 127, Metal 009, Organ 014, Consonant 033, Vowel Bass 035, Consonant 001, WaveEdit Iterativ, Consonant 129, Tube 066, Organ 025, Vowel Tenor 012, Chord 039, Sampled 106, Sampled 207, Breath 013, WaveEdit Grav_c2, Consonant 079, Otmorph 119, Consonant 064, WaveEdit Ppg_wa18, WaveEdit Pd103, Overtone 033, Chord 013, Vowel Bass 014, Sampled 226, Chord 011, Sampled 048, Vowel Tenor 037, Otmorph 095, Organ 012, Vowel Tenor 019, Overtone 015, Overtone 054, Consonant 088, Consonant 108, Sampled 197, Organ 037, Choir 004, Vowel Bass 030, Resonator 019, Otmorph 086, Tube 074, Tube 089, Sampled 182, Organ 038, Sampled 173, Consonant 060, Otmorph 030, Sampled 138, Consonant 056, Otmorph 132, Otmorph 138, Glass 003, Vowel Bass 021, Vowel Tenor 036, Organ 033, Sampled 214, Consonant 081, Consonant 094, Consonant 028, Consonant 003, Vowel Soprano 035, Consonant 126, Vowel Bass 027, Chord 000, Resonator 010, Choir 026, Metal 015, AKWF 0018-02, Consonant 046, Otmorph 079, Vowel Tenor 024, Consonant 117, Choir 025, Overtone 067, Chord 016, Otmorph 109, Vowel Alto 001, Otmorph 089, Consonant 077, Shimmer 033, Otmorph 022, Consonant 144, Organ 000, Consonant 139, Consonant 106, Vowel Bass 039, Otmorph 043, Vowel Tenor 033, Resonator 012, Otmorph 059, Otmorph 073, Sampled 093, Overtone 055, Sampled 189, Otmorph 068, Otmorph 064, Breath 029, Chord 033, Sampled 108, Consonant 030, Bowed 075, Organ 011, Vowel Tenor 006, Otmorph 056, AKWF 0006-02, Glass 037, Vowel Bass 001, WaveEdit Tidyb026, WaveEdit Sohler39, Sampled 076, Organ 021, WaveEdit Harmonio, WaveEdit Grav_c9, Consonant 037, WaveEdit Sine_a02, WaveEdit High_fre, WaveEdit Tidyb024, AKWF hvoice-01, WaveEdit Sohler58, WaveEdit Ppg_wa29, Vowel Bass 036, WaveEdit Sohler19, WaveEdit Sohler53, WaveEdit Sohler94, Glass 023, Chord 031, WaveEdit Access_v, Bowed 079, AKWF c604-01, WaveEdit Sohler56, WaveEdit Tidyb017, WaveEdit Sohler11, Overtone 018, Consonant 032, Reed 034, WaveEdit Mk_dwg_h, WaveEdit Grav_c3, WaveEdit Fracta01, WaveEdit Synlp54, WaveEdit Tidyb062, WaveEdit Envelo01, Vowel Tenor 005, WaveEdit Tidyb004, Chord 008, AKWF bitreduced-01, AKWF bw-saw-01, Overtone 053, WaveEdit Grav_a6, WaveEdit Sohler26, Sampled 066, WaveEdit Sohler86, Tube 036, Otmorph 014, WaveEdit Sohler76, WaveEdit Sohler27, Choir 030, WaveEdit Bank_410, WaveEdit Tidyb018, WaveEdit Alpha_2, WaveEdit Tidyb003, WaveEdit Sohler35, WaveEdit Lfo_play, WaveEdit Ppg_wa01, WaveEdit Ppg_wa02, WaveEdit Prophe00, Vowel Tenor 011, Tube 072, WaveEdit Tidyb058, Consonant 137, WaveEdit Sohler31, Consonant 076, WaveEdit Just_ran, Consonant 080, WaveEdit Grav_a5, WaveEdit Sohler14, Resonator 020, WaveEdit Grav_a1, Glass 020, WaveEdit Sohler97, Chord 028, WaveEdit Grav_a2, Vowel Bass 034, WaveEdit Sohler70, Breath 031, AKWF oscchip-03, WaveEdit Sohler84, Consonant 149, WaveEdit Tidyb041, Consonant 023, Otmorph 126, Bowed 054, WaveEdit Mixed_as, Otmorph 084, WaveEdit Twist31, WaveEdit Spectr05, WaveEdit Sohler92, WaveEdit Isobelle, Chord 021, Consonant 067, WaveEdit Envelope, Sub 003, WaveEdit Sohler79, Tube 002, Consonant 008, WaveEdit Ppg_wa03, Pluck 028, Vowel Alto 012, AKWF epiano-01, AKWF bw-squ-02, Organ 005, WaveEdit Sine2saw, Vowel Alto 039, Glass 004, WaveEdit Sohler43, Sampled 109, Vowel Alto 003, Glass 010, AKWF clarinett-01, WaveEdit Sohler18, Sampled 170, Tube 046,

Vowel Tenor 017, WaveEdit Foldfeed, Tube 008, AKWF 0015-01, Otmorph 106, Overtone 037, Consonant 022, Breath 019, WaveEdit Ppg_wa07, Sampled 164, WaveEdit Sine_mut, Vowel Alto 007, Sampled 211, Metal 025, Glass 007, Vowel Soprano 011, Glass 024, AKWF 0008-01, Vowel Soprano 033, Shimmer 002, Pure 015, WaveEdit Geometri, Sampled 203, Chord 007, Vowel Soprano 005, AKWF altosax-01, Vowel Soprano 015, WaveEdit Ppg_wa17, Overtone 010, WaveEdit Synlp159, Sampled 115, WaveEdit Ppg_wa22, Otmorph 112, Vowel Alto 016, Breath 033, Tube 009, Tube 079, Sampled 097, Chord 023, Stack 008, AKWF 0003-01, Glass 038, Metal 016

Position	arp.position	0 .. 1	0.3
Position Env	arp.pos_env	-1 .. 1	0
Position Decay	arp.pos_decay	10 .. 8000 ms	1500 ms
Position LFO	arp.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	arp.pos_lfo_beats	0.25 .. 64 beats	16 beats
Cutoff	arp.cutoff	200 .. 18000 Hz	2600 Hz
Resonance	arp.resonance	0 .. 1	0.15
Env Amount	arp.env_amount	0 .. 6 oct	2.5 oct
Filter Decay	arp.filter_decay	5 .. 3000 ms	120 ms
Key Track	arp.key_track	0 .. 1	0.5
HP Floor	arp.hp_floor	40 .. 400 Hz	300 Hz
HP Track	arp.hp_track	0 .. 1 x f0	1 x f0
Attack	arp.amp_attack	0.3 .. 2000 ms	0.3 ms
Decay	arp.amp_decay	5 .. 4000 ms	150 ms
Sustain	arp.amp_sustain	0 .. 1	0
Release	arp.amp_release	5 .. 4000 ms	60 ms
Width	arp.width	0 .. 1	0.8
Velocity	arp.vel_sens	0 .. 1	0.4
Delay Send	arp.delay_send	0 .. 1	0.2
Delay Left	arp.delay_left	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/8
Delay Right	arp.delay_right	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Delay Feedback	arp.delay_feedback	0 .. 0.9	0.3
Delay High Pass	arp.delay_high_pass	150 .. 2000 Hz	350 Hz
Delay Low Pass	arp.delay_low_pass	800 .. 16000 Hz	6000 Hz

Room Send	arp.room_send	0 .. 1	0.15
Hall Send	arp.hall_send	0 .. 1	0.1
Duck	arp.duck	0 .. 18 dB	3 dB
Trance Gate	arp.gate	off / on	off
Gate Pattern	arp.gate_pattern	Sixteenths, Eighths, Rolling, Gallop, 3-3-2, Triplets, Trance 1, Trance 2, Trance 3	Trance 1
Gate Depth	arp.gate_depth	0 .. 1	0.85
Gate Duty	arp.gate_duty	0.05 .. 1	0.5
Gate Attack	arp.gate_attack	0.5 .. 60 ms	3 ms
Gate Release	arp.gate_release	0.5 .. 120 ms	12 ms
Gate Tone	arp.gate_tone	0 .. 1	0.4
Level	arp.level	-36 .. 6 dB	-11 dB
Disperse	arp.disperse	0 .. 8 x	0 x
Disperse Freq	arp.disperse_freq	200 .. 8000 Hz	1250 Hz
Drift	arp.drift	0 .. 8 ct	1 ct
Filter Type	arp.filter_type	Low Pass, Band Pass, High Pass, Notch	Low Pass
Glide	arp.glide	0 .. 400 ms	0 ms
Pan	arp.pan	-1 .. 1	0.15
Oscillator 2	arp.osc2	Off, Supersaw, VA, FM, Wavetable	Off
Osc 2 Mix	arp.osc2_mix	0 .. 1	0
Osc 2 Interval	arp.osc2_interval	-2 Oct, -1 Oct, -5th, Unison, +5th, +1 Oct	-1 Oct
Osc 2 Detune	arp.osc2_detune	-50 .. 50 ct	0 ct
Voice LFO	arp.lfo_beats	0.25 .. 64 beats	8 beats
LFO to Cutoff	arp.lfo_cutoff	0 .. 4 oct	0 oct
LFO to Pitch	arp.lfo_pitch	0 .. 50 ct	0 ct
LFO to Level	arp.lfo_amp	0 .. 1	0
Plate Send	arp.plate_send	0 .. 1	0
Slow Movement	arp.slow_mod	0 .. 1	0
Filter Model	arp.filter_model	State Variable, Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder, Korg35 (MS-20), Polivoks, Wasp	State Variable

Filter Mode	arp.filter_mode	0 .. 1	0
Filter Attack	arp.filt_attack	0.1 .. 4000 ms	0.1 ms
Filter Sustain	arp.filt_sustain	0 .. 1	0
Filter Release	arp.filt_release	5 .. 8000 ms	300 ms
Mod Attack	arp.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	arp.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	arp.menv_sustain	0 .. 1	0
Mod Release	arp.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	arp.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	arp.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	arp.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	arp.lfo1_retrig	off / on	off
LFO 1 Fade	arp.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	arp.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	arp.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	arp.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	arp.lfo2_retrig	off / on	off
LFO 2 Fade	arp.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	arp.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	arp.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	arp.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	arp.lfo3_retrig	off / on	off
LFO 3 Fade	arp.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	arp.lfo4_rate	0.01 .. 40 Hz	1 Hz

LFO 4 Shape	arp.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	arp.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	arp.lfo4_retrig	off / on	off
LFO 4 Fade	arp.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	arp.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	arp.mx1_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 1 Amount	arp.mx1_amount	-1 .. 1	0
Mod 2 Source	arp.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	arp.mx2_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 2 Amount	arp.mx2_amount	-1 .. 1	0
Mod 3 Source	arp.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	arp.mx3_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 3 Amount	arp.mx3_amount	-1 .. 1	0
Mod 4 Source	arp.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	arp.mx4_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 4 Amount	arp.mx4_amount	-1 .. 1	0
Mod 5 Source	arp.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	arp.mx5_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 5 Amount	arp.mx5_amount	-1 .. 1	0
Mod 6 Source	arp.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	arp.mx6_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 6 Amount	arp.mx6_amount	-1 .. 1	0
Mod 7 Source	arp.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off

Mod 7 Target	arp.mx7_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 7 Amount	arp.mx7_amount	-1 .. 1	0
Mod 8 Source	arp.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	arp.mx8_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 8 Amount	arp.mx8_amount	-1 .. 1	0

PAD

Control	Key	Range	Default
Oscillator	pad.osc	Supersaw, VA, FM, Wavetable	Supersaw
Detune	pad.detune	0 .. 1	0.5
Mix	pad.mix	0 .. 1	0.85
Dynamic Detune	pad.dynamic_detune	0 .. 1	0
Wave	pad.wave	0 .. 1	0
Pulse Width	pad.pulse_width	0.05 .. 0.95	0.5
FM Ratio	pad.fm_ratio	0.5 .. 8	2
FM Index	pad.fm_index	0 .. 10	2.5
FM Decay	pad.fm_decay	5 .. 2000 ms	250 ms
Table	pad.table	Classic, Vocal, Glass, PWM, Sync, Formant Saw, WaveEdit Hienharm, WaveEdit Junox_ho, WaveEdit Euclidean, WaveEdit Sohler49, WaveEdit Vocal_fo, WaveEdit Pwn_saw, WaveEdit Tidyb030, WaveEdit I_heart, WaveEdit Sine_n, WaveEdit Vocal_00, WaveEdit Supersaw, WaveEdit Tidyb060, Consonant 024, WaveEdit Synlp92, WaveEdit Distorte, WaveEdit Fourier2, WaveEdit Grav_b8, WaveEdit Spectr01, WaveEdit Discorda, WaveEdit Sohler47, WaveEdit Ppg_wa27, WaveEdit Micro_q, WaveEdit Noise_wa, WaveEdit Kuato, WaveEdit Synlp67, WaveEdit Komple01, Consonant 066, WaveEdit Iterat00, WaveEdit Retro_sp, WaveEdit Acid_rin, WaveEdit Meravigl, WaveEdit Tidyb066, WaveEdit Tidyb049, WaveEdit Fracta03, WaveEdit Chebyshe, Sampled 167, WaveEdit Wavetabl, Chord 004, WaveEdit Grav_b6, WaveEdit Sohler33, WaveEdit Ppg_wa13, WaveEdit Pd104, WaveEdit Sohler48, WaveEdit Ppg_wa28, WaveEdit Wavetrip, WaveEdit Cz_ish, WaveEdit Fx_bitno, WaveEdit Tidyb046, WaveEdit Sohler74, WaveEdit Sohler29, WaveEdit Enshtu02, WaveEdit Tidyb072, WaveEdit Tidyb050, WaveEdit Voice_a, Resonator 018, WaveEdit Tidyb035, WaveEdit Tezzalog, AKWF 0004-hollow-01, WaveEdit Lom_a, WaveEdit Tidyb064, WaveEdit Moddrop, WaveEdit Harmomet, Hollow 013, WaveEdit Tidyb065, WaveEdit Ultra_ch, WaveEdit Laser_cr, WaveEdit Ppg_wa05, WaveEdit Drone, WaveEdit Gentle_m, WaveEdit Tidyb067, WaveEdit Tidyb005, Resonator 008, WaveEdit Synlp71, WaveEdit Sohler46, WaveEdit Tidyb077, WaveEdit Tidyb027, WaveEdit Sand_eye, WaveEdit Feedback, WaveEdit Tidyb071, WaveEdit Snake_ey, WaveEdit Hyperbol, AKWF 0018-hollow-01, Overtone 064, WaveEdit Light_00, WaveEdit Crush_ad, Organ 034, Sampled 131, Glass 012, WaveEdit Sohler67, Organ 006, WaveEdit Ppg_wa14, WaveEdit Sq8_sh, AKWF clavinet-hollow-01, Glass 009, WaveEdit Grav_c7,	Vocal

Consonant 100, WaveEdit Sohler51, Vowel Bass 015, Sampled 227, Hollow 005, Sampled 065, AKWF oscchip-04, AKWF raw-01, Hollow 015, AKWF snippets-01, Chord 027, Consonant 071, Chord 020, Consonant 051, WaveEdit Cyborg, Sampled 238, AKWF 0014-hollow-01, Overtone 000, WaveEdit Ppg_wa15, WaveEdit Grav_b4, Consonant 009, WaveEdit Grav_c5, WaveEdit Twist6, Glass 027, Sampled 078, AKWF oscchip-05, Consonant 120, WaveEdit Ppg_wa04, WaveEdit Sohler59, Sampled 081, Otmorph 053, Chord 030, Chord 032, Organ 029, Overtone 108, Hollow 001, WaveEdit Ppg_wa10, WaveEdit Organ_di, AKWF 0009-hollow-01, Tube 042, Vowel Bass 008, Organ 035, Metal 001, Consonant 114, AKWF vgame-03, Sampled 224, Resonator 001, Vowel Bass 038, Sampled 058, Vowel Tenor 038, Glass 026, Chord 019, Otmorph 115, AKWF stereo-04, Consonant 105, Vowel Bass 026, Chord 017, WaveEdit Ppg_wa11, Consonant 109, Sampled 125, Consonant 012, Overtone 046, Glass 016, Otmorph 067, Organ 026, Otmorph 005, Glass 018, Consonant 018, Hollow 023, Overtone 019, Otmorph 070, Consonant 127, Vowel Bass 029, Organ 016, Vowel Bass 017, Sampled 209, Sampled 130, Overtone 038, WaveEdit Reso_ana, Consonant 142, WaveEdit Reso_p00, Chord 009, AKWF distorted-01, Sampled 162, Glass 006, Metal 003, WaveEdit Square_p, Sampled 059, Organ 030, Sampled 003, Consonant 036, Chord 034, Sampled 193, WaveEdit Qux_fmy, Vowel Bass 023, Otmorph 133, WaveEdit Sohler52, Vowel Bass 007, WaveEdit Grav_b3, Consonant 025, Consonant 146, Chord 018, Overtone 047, Chord 038, Sampled 127, Metal 009, Organ 014, Consonant 033, Vowel Bass 035, Consonant 001, WaveEdit Iterativ, Consonant 129, Tube 066, Organ 025, Vowel Tenor 012, Chord 039, Sampled 106, Sampled 207, Breath 013, WaveEdit Grav_c2, Consonant 079, Otmorph 119, Consonant 064, WaveEdit Ppg_wa18, WaveEdit Pd103, Overtone 033, Chord 013, Vowel Bass 014, Sampled 226, Chord 011, Sampled 048, Vowel Tenor 037, Otmorph 095, Organ 012, Vowel Tenor 019, Overtone 015, Overtone 054, Consonant 088, Consonant 108, Sampled 197, Organ 037, Choir 004, Vowel Bass 030, Resonator 019, Otmorph 086, Tube 074, Tube 089, Sampled 182, Organ 038, Sampled 173, Consonant 060, Otmorph 030, Sampled 138, Consonant 056, Otmorph 132, Otmorph 138, Glass 003, Vowel Bass 021, Vowel Tenor 036, Organ 033, Sampled 214, Consonant 081, Consonant 094, Consonant 028, Consonant 003, Vowel Soprano 035, Consonant 126, Vowel Bass 027, Chord 000, Resonator 010, Choir 026, Metal 015, AKWF 0018-02, Consonant 046, Otmorph 079, Vowel Tenor 024, Consonant 117, Choir 025, Overtone 067, Chord 016, Otmorph 109, Vowel Alto 001, Otmorph 089, Consonant 077, Shimmer 033, Otmorph 022, Consonant 144, Organ 000, Consonant 139, Consonant 106, Vowel Bass 039, Otmorph 043, Vowel Tenor 033, Resonator 012, Otmorph 059, Otmorph 073, Sampled 093, Overtone 055, Sampled 189, Otmorph 068, Otmorph 064, Breath 029, Chord 033, Sampled 108, Consonant 030, Bowed 075, Organ 011, Vowel Tenor 006, Otmorph 056, AKWF 0006-02, Glass 037, Vowel Bass 001, WaveEdit Tidyb026, WaveEdit Sohler39, Sampled 076, Organ 021, WaveEdit Harmonio, WaveEdit Grav_c9, Consonant 037, WaveEdit Sine_a02, WaveEdit High_fre, WaveEdit Tidyb024, AKWF hvoice-01, WaveEdit Sohler58, WaveEdit Ppg_wa29, Vowel Bass 036, WaveEdit Sohler19, WaveEdit Sohler53, WaveEdit Sohler94, Glass 023, Chord 031, WaveEdit Access_v, Bowed 079, AKWF c604-01, WaveEdit Sohler56, WaveEdit Tidyb017, WaveEdit Sohler11, Overtone 018, Consonant 032, Reed 034, WaveEdit Mk_dwg_h, WaveEdit Grav_c3, WaveEdit Fracta01, WaveEdit Synlp54, WaveEdit Tidyb062, WaveEdit Envelo01, Vowel Tenor 005, WaveEdit Tidyb004, Chord 008, AKWF bitreduced-01, AKWF bw-saw-01, Overtone 053, WaveEdit Grav_a6, WaveEdit Sohler26, Sampled 066, WaveEdit Sohler86, Tube 036, Otmorph 014, WaveEdit Sohler76, WaveEdit Sohler27, Choir 030, WaveEdit Bank_410, WaveEdit Tidyb018, WaveEdit Alpha_2, WaveEdit Tidyb003, WaveEdit Sohler35, WaveEdit Lfo_play, WaveEdit Ppg_wa01, WaveEdit Ppg_wa02, WaveEdit Prophe00, Vowel Tenor 011, Tube 072, WaveEdit

Tidyb058, Consonant 137, WaveEdit Sohler31, Consonant 076, WaveEdit Just_ran, Consonant 080, WaveEdit Grav_a5, WaveEdit Sohler14, Resonator 020, WaveEdit Grav_a1, Glass 020, WaveEdit Sohler97, Chord 028, WaveEdit Grav_a2, Vowel Bass 034, WaveEdit Sohler70, Breath 031, AKWF oscchip-03, WaveEdit Sohler84, Consonant 149, WaveEdit Tidyb041, Consonant 023, Otmorph 126, Bowed 054, WaveEdit Mixed_as, Otmorph 084, WaveEdit Twist31, WaveEdit Spectr05, WaveEdit Sohler92, WaveEdit Isobelle, Chord 021, Consonant 067, WaveEdit Envelope, Sub 003, WaveEdit Sohler79, Tube 002, Consonant 008, WaveEdit Ppg_wa03, Pluck 028, Vowel Alto 012, AKWF epiano-01, AKWF bw-squ-02, Organ 005, WaveEdit Sine2saw, Vowel Alto 039, Glass 004, WaveEdit Sohler43, Sampled 109, Vowel Alto 003, Glass 010, AKWF clarinett-01, WaveEdit Sohler18, Sampled 170, Tube 046, Vowel Tenor 017, WaveEdit Foldfeed, Tube 008, AKWF 0015-01, Otmorph 106, Overtone 037, Consonant 022, Breath 019, WaveEdit Ppg_wa07, Sampled 164, WaveEdit Sine_mut, Vowel Alto 007, Sampled 211, Metal 025, Glass 007, Vowel Soprano 011, Glass 024, AKWF 0008-01, Vowel Soprano 033, Shimmer 002, Pure 015, WaveEdit Geometri, Sampled 203, Chord 007, Vowel Soprano 005, AKWF altosax-01, Vowel Soprano 015, WaveEdit Ppg_wa17, Overtone 010, WaveEdit Synlp159, Sampled 115, WaveEdit Ppg_wa22, Otmorph 112, Vowel Alto 016, Breath 033, Tube 009, Tube 079, Sampled 097, Chord 023, Stack 008, AKWF 0003-01, Glass 038, Metal 016

Position	pad.position	0 .. 1	0.3
Position Env	pad.pos_env	-1 .. 1	0
Position Decay	pad.pos_decay	10 .. 8000 ms	1500 ms
Position LFO	pad.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	pad.pos_lfo_beats	0.25 .. 64 beats	16 beats
Cutoff	pad.cutoff	200 .. 18000 Hz	2200 Hz
Resonance	pad.resonance	0 .. 1	0.1
Env Amount	pad.env_amount	0 .. 6 oct	0.5 oct
Filter Decay	pad.filter_decay	5 .. 3000 ms	2000 ms
Key Track	pad.key_track	0 .. 1	0.5
HP Floor	pad.hp_floor	40 .. 400 Hz	200 Hz
HP Track	pad.hp_track	0 .. 1 x f0	0.6 x f0
Attack	pad.amp_attack	0.3 .. 2000 ms	800 ms
Decay	pad.amp_decay	5 .. 4000 ms	2000 ms
Sustain	pad.amp_sustain	0 .. 1	0.9
Release	pad.amp_release	5 .. 4000 ms	1500 ms
Width	pad.width	0 .. 1	1
Velocity	pad.vel_sens	0 .. 1	0.4
Delay Send	pad.delay_send	0 .. 1	0
Delay Left	pad.delay_left	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16

Delay Right	pad.delay_right	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	1/4
Delay Feedback	pad.delay_feedback	0 .. 0.9	0.4
Delay High Pass	pad.delay_high_pass	150 .. 2000 Hz	350 Hz
Delay Low Pass	pad.delay_low_pass	800 .. 16000 Hz	6000 Hz
Room Send	pad.room_send	0 .. 1	0
Hall Send	pad.hall_send	0 .. 1	0.45
Duck	pad.duck	0 .. 18 dB	4 dB
Trance Gate	pad.gate	off / on	off
Gate Pattern	pad.gate_pattern	Sixteenths, Eighths, Rolling, Gallop, 3-3-2, Triplets, Trance 1, Trance 2, Trance 3	Trance 1
Gate Depth	pad.gate_depth	0 .. 1	0.85
Gate Duty	pad.gate_duty	0.05 .. 1	0.5
Gate Attack	pad.gate_attack	0.5 .. 60 ms	3 ms
Gate Release	pad.gate_release	0.5 .. 120 ms	12 ms
Gate Tone	pad.gate_tone	0 .. 1	0.4
Level	pad.level	-36 .. 6 dB	-17 dB
Disperse	pad.disperse	0 .. 8 x	0 x
Disperse Freq	pad.disperse_freq	200 .. 8000 Hz	1250 Hz
Drift	pad.drift	0 .. 8 ct	1 ct
Filter Type	pad.filter_type	Low Pass, Band Pass, High Pass, Notch	Low Pass
Glide	pad.glide	0 .. 400 ms	0 ms
Pan	pad.pan	-1 .. 1	0
Oscillator 2	pad.osc2	Off, Supersaw, VA, FM, Wavetable	Off
Osc 2 Mix	pad.osc2_mix	0 .. 1	0
Osc 2 Interval	pad.osc2_interval	-2 Oct, -1 Oct, -5th, Unison, +5th, +1 Oct	-1 Oct
Osc 2 Detune	pad.osc2_detune	-50 .. 50 ct	0 ct
Voice LFO	pad.lfo_beats	0.25 .. 64 beats	16 beats
LFO to Cutoff	pad.lfo_cutoff	0 .. 4 oct	0.3 oct
LFO to Pitch	pad.lfo_pitch	0 .. 50 ct	0 ct

LFO to Level	pad.lfo_amp	0 .. 1	0
Plate Send	pad.plate_send	0 .. 1	0.1
Slow Movement	pad.slow_mod	0 .. 1	0
Filter Model	pad.filter_model	State Variable, Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder, Korg35 (MS-20), Polivoks, Wasp	State Variable
Filter Mode	pad.filter_mode	0 .. 1	0
Filter Attack	pad.filt_attack	0.1 .. 4000 ms	0.1 ms
Filter Sustain	pad.filt_sustain	0 .. 1	0
Filter Release	pad.filt_release	5 .. 8000 ms	300 ms
Mod Attack	pad.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	pad.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	pad.menv_sustain	0 .. 1	0
Mod Release	pad.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	pad.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	pad.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	pad.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrig	pad.lfo1_retrig	off / on	off
LFO 1 Fade	pad.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	pad.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	pad.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	pad.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrig	pad.lfo2_retrig	off / on	off
LFO 2 Fade	pad.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	pad.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	pad.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine

LFO 3 Sync	pad.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	pad.lfo3_retrig	off / on	off
LFO 3 Fade	pad.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	pad.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	pad.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	pad.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	pad.lfo4_retrig	off / on	off
LFO 4 Fade	pad.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	pad.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	pad.mx1_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 1 Amount	pad.mx1_amount	-1 .. 1	0
Mod 2 Source	pad.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	pad.mx2_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 2 Amount	pad.mx2_amount	-1 .. 1	0
Mod 3 Source	pad.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	pad.mx3_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 3 Amount	pad.mx3_amount	-1 .. 1	0
Mod 4 Source	pad.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	pad.mx4_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 4 Amount	pad.mx4_amount	-1 .. 1	0
Mod 5 Source	pad.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	pad.mx5_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 5 Amount	pad.mx5_amount	-1 .. 1	0

Mod 6 Source	pad.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	pad.mx6_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 6 Amount	pad.mx6_amount	-1 .. 1	0
Mod 7 Source	pad.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	pad.mx7_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 7 Amount	pad.mx7_amount	-1 .. 1	0
Mod 8 Source	pad.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	pad.mx8_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 8 Amount	pad.mx8_amount	-1 .. 1	0

STAB

Control	Key	Range	Default
Oscillator	stab.osc	Supersaw, VA, FM, Wavetable	Supersaw
Detune	stab.detune	0 .. 1	0.4
Mix	stab.mix	0 .. 1	0.7
Dynamic Detune	stab.dynamic_detune	0 .. 1	0.7
Wave	stab.wave	0 .. 1	0
Pulse Width	stab.pulse_width	0.05 .. 0.95	0.5
FM Ratio	stab.fm_ratio	0.5 .. 8	2
FM Index	stab.fm_index	0 .. 10	2.5
FM Decay	stab.fm_decay	5 .. 2000 ms	250 ms
Table	stab.table	Classic, Vocal, Glass, PWM, Sync, Formant Saw, WaveEdit Hienharm, WaveEdit Junox_ho, WaveEdit Euclidea, WaveEdit Sohler49, WaveEdit Vocal_fo, WaveEdit Pwn_saw, WaveEdit Tidyb030, WaveEdit I_heart, WaveEdit Sine_n, WaveEdit Vocal_00, WaveEdit Supersaw, WaveEdit Tidyb060, Consonant 024, WaveEdit Synlp92, WaveEdit Distorte, WaveEdit Fourier2, WaveEdit Grav_b8, WaveEdit Spectr01, WaveEdit Discorda, WaveEdit Sohler47, WaveEdit Ppg_wa27, WaveEdit Micro_q, WaveEdit Noise_wa, WaveEdit Kuato, WaveEdit Synlp67, WaveEdit Komple01, Consonant 066, WaveEdit Iterat00, WaveEdit Retro_sp, WaveEdit Acid_rin, WaveEdit Meravigl, WaveEdit Tidyb066, WaveEdit Tidyb049, WaveEdit Fracta03, WaveEdit Chebyshe, Sampled 167, WaveEdit Wavetabl, Chord 004, WaveEdit Grav_b6, WaveEdit Sohler33, WaveEdit Ppg_wa13, WaveEdit Pd104, WaveEdit Sohler48, WaveEdit Ppg_wa28, WaveEdit Wavetrip, WaveEdit Cz_ish, WaveEdit Fx_bitno, WaveEdit Tidyb046, WaveEdit Sohler74, WaveEdit Sohler29, WaveEdit Enshtu02, WaveEdit Tidyb072, WaveEdit	Vocal

Tidyb050, WaveEdit Voice_a, Resonator 018, WaveEdit Tidyb035, WaveEdit Tezzalog, AKWF 0004-hollow-01, WaveEdit Lom_a, WaveEdit Tidyb064, WaveEdit Moddrop, WaveEdit Harmomet, Hollow 013, WaveEdit Tidyb065, WaveEdit Ultra_ch, WaveEdit Laser_cr, WaveEdit Ppg_wa05, WaveEdit Drone, WaveEdit Gentle_m, WaveEdit Tidyb067, WaveEdit Tidyb005, Resonator 008, WaveEdit Synlp71, WaveEdit Sohler46, WaveEdit Tidyb077, WaveEdit Tidyb027, WaveEdit Sand_eye, WaveEdit Feedback, WaveEdit Tidyb071, WaveEdit Snake_ey, WaveEdit Hyperbol, AKWF 0018-hollow-01, Overtone 064, WaveEdit Light_00, WaveEdit Crush_ad, Organ 034, Sampled 131, Glass 012, WaveEdit Sohler67, Organ 006, WaveEdit Ppg_wa14, WaveEdit Sq8_sh, AKWF clavinet-hollow-01, Glass 009, WaveEdit Grav_c7, Consonant 100, WaveEdit Sohler51, Vowel Bass 015, Sampled 227, Hollow 005, Sampled 065, AKWF oscchip-04, AKWF raw-01, Hollow 015, AKWF snippets-01, Chord 027, Consonant 071, Chord 020, Consonant 051, WaveEdit Cyborg, Sampled 238, AKWF 0014-hollow-01, Overtone 000, WaveEdit Ppg_wa15, WaveEdit Grav_b4, Consonant 009, WaveEdit Grav_c5, WaveEdit Twist6, Glass 027, Sampled 078, AKWF oscchip-05, Consonant 120, WaveEdit Ppg_wa04, WaveEdit Sohler59, Sampled 081, Otmorph 053, Chord 030, Chord 032, Organ 029, Overtone 108, Hollow 001, WaveEdit Ppg_wa10, WaveEdit Organ_di, AKWF 0009-hollow-01, Tube 042, Vowel Bass 008, Organ 035, Metal 001, Consonant 114, AKWF vgame-03, Sampled 224, Resonator 001, Vowel Bass 038, Sampled 058, Vowel Tenor 038, Glass 026, Chord 019, Otmorph 115, AKWF stereo-04, Consonant 105, Vowel Bass 026, Chord 017, WaveEdit Ppg_wa11, Consonant 109, Sampled 125, Consonant 012, Overtone 046, Glass 016, Otmorph 067, Organ 026, Otmorph 005, Glass 018, Consonant 018, Hollow 023, Overtone 019, Otmorph 070, Consonant 127, Vowel Bass 029, Organ 016, Vowel Bass 017, Sampled 209, Sampled 130, Overtone 038, WaveEdit Reso_ana, Consonant 142, WaveEdit Reso_p00, Chord 009, AKWF distorted-01, Sampled 162, Glass 006, Metal 003, WaveEdit Square_p, Sampled 059, Organ 030, Sampled 003, Consonant 036, Chord 034, Sampled 193, WaveEdit Qux_fmy, Vowel Bass 023, Otmorph 133, WaveEdit Sohler52, Vowel Bass 007, WaveEdit Grav_b3, Consonant 025, Consonant 146, Chord 018, Overtone 047, Chord 038, Sampled 127, Metal 009, Organ 014, Consonant 033, Vowel Bass 035, Consonant 001, WaveEdit Iterativ, Consonant 129, Tube 066, Organ 025, Vowel Tenor 012, Chord 039, Sampled 106, Sampled 207, Breath 013, WaveEdit Grav_c2, Consonant 079, Otmorph 119, Consonant 064, WaveEdit Ppg_wa18, WaveEdit Pd103, Overtone 033, Chord 013, Vowel Bass 014, Sampled 226, Chord 011, Sampled 048, Vowel Tenor 037, Otmorph 095, Organ 012, Vowel Tenor 019, Overtone 015, Overtone 054, Consonant 088, Consonant 108, Sampled 197, Organ 037, Choir 004, Vowel Bass 030, Resonator 019, Otmorph 086, Tube 074, Tube 089, Sampled 182, Organ 038, Sampled 173, Consonant 060, Otmorph 030, Sampled 138, Consonant 056, Otmorph 132, Otmorph 138, Glass 003, Vowel Bass 021, Vowel Tenor 036, Organ 033, Sampled 214, Consonant 081, Consonant 094, Consonant 028, Consonant 003, Vowel Soprano 035, Consonant 126, Vowel Bass 027, Chord 000, Resonator 010, Choir 026, Metal 015, AKWF 0018-02, Consonant 046, Otmorph 079, Vowel Tenor 024, Consonant 117, Choir 025, Overtone 067, Chord 016, Otmorph 109, Vowel Alto 001, Otmorph 089, Consonant 077, Shimmer 033, Otmorph 022, Consonant 144, Organ 000, Consonant 139, Consonant 106, Vowel Bass 039, Otmorph 043, Vowel Tenor 033, Resonator 012, Otmorph 059, Otmorph 073, Sampled 093, Overtone 055, Sampled 189, Otmorph 068, Otmorph 064, Breath 029, Chord 033, Sampled 108, Consonant 030, Bowed 075, Organ 011, Vowel Tenor 006, Otmorph 056, AKWF 0006-02, Glass 037, Vowel Bass 001, WaveEdit Tidyb026, WaveEdit Sohler39, Sampled 076, Organ 021, WaveEdit Harmonio, WaveEdit Grav_c9, Consonant 037, WaveEdit Sine_a02, WaveEdit High_fre, WaveEdit Tidyb024,

AKWF hvoice-01, WaveEdit Sohler58, WaveEdit Ppg_wa29, Vowel Bass 036, WaveEdit Sohler19, WaveEdit Sohler53, WaveEdit Sohler94, Glass 023, Chord 031, WaveEdit Access_v, Bowed 079, AKWF c604-01, WaveEdit Sohler56, WaveEdit Tidyb017, WaveEdit Sohler11, Overtone 018, Consonant 032, Reed 034, WaveEdit Mk_dwg_h, WaveEdit Grav_c3, WaveEdit Fracta01, WaveEdit Synlp54, WaveEdit Tidyb062, WaveEdit Envelo01, Vowel Tenor 005, WaveEdit Tidyb004, Chord 008, AKWF bitreduced-01, AKWF bw-saw-01, Overtone 053, WaveEdit Grav_a6, WaveEdit Sohler26, Sampled 066, WaveEdit Sohler86, Tube 036, Otmorph 014, WaveEdit Sohler76, WaveEdit Sohler27, Choir 030, WaveEdit Bank_410, WaveEdit Tidyb018, WaveEdit Alpha_2, WaveEdit Tidyb003, WaveEdit Sohler35, WaveEdit Lfo_play, WaveEdit Ppg_wa01, WaveEdit Ppg_wa02, WaveEdit Prophe00, Vowel Tenor 011, Tube 072, WaveEdit Tidyb058, Consonant 137, WaveEdit Sohler31, Consonant 076, WaveEdit Just_ran, Consonant 080, WaveEdit Grav_a5, WaveEdit Sohler14, Resonator 020, WaveEdit Grav_a1, Glass 020, WaveEdit Sohler97, Chord 028, WaveEdit Grav_a2, Vowel Bass 034, WaveEdit Sohler70, Breath 031, AKWF oscchip-03, WaveEdit Sohler84, Consonant 149, WaveEdit Tidyb041, Consonant 023, Otmorph 126, Bowed 054, WaveEdit Mixed_as, Otmorph 084, WaveEdit Twist31, WaveEdit Spectr05, WaveEdit Sohler92, WaveEdit Isobelle, Chord 021, Consonant 067, WaveEdit Envelope, Sub 003, WaveEdit Sohler79, Tube 002, Consonant 008, WaveEdit Ppg_wa03, Pluck 028, Vowel Alto 012, AKWF epiano-01, AKWF bw-squ-02, Organ 005, WaveEdit Sine2saw, Vowel Alto 039, Glass 004, WaveEdit Sohler43, Sampled 109, Vowel Alto 003, Glass 010, AKWF clarinett-01, WaveEdit Sohler18, Sampled 170, Tube 046, Vowel Tenor 017, WaveEdit Foldfeed, Tube 008, AKWF 0015-01, Otmorph 106, Overtone 037, Consonant 022, Breath 019, WaveEdit Ppg_wa07, Sampled 164, WaveEdit Sine_mut, Vowel Alto 007, Sampled 211, Metal 025, Glass 007, Vowel Soprano 011, Glass 024, AKWF 0008-01, Vowel Soprano 033, Shimmer 002, Pure 015, WaveEdit Geometri, Sampled 203, Chord 007, Vowel Soprano 005, AKWF altosax-01, Vowel Soprano 015, WaveEdit Ppg_wa17, Overtone 010, WaveEdit Synlp159, Sampled 115, WaveEdit Ppg_wa22, Otmorph 112, Vowel Alto 016, Breath 033, Tube 009, Tube 079, Sampled 097, Chord 023, Stack 008, AKWF 0003-01, Glass 038, Metal 016

Position	stab.position	0 .. 1	0.3
Position Env	stab.pos_env	-1 .. 1	0
Position Decay	stab.pos_decay	10 .. 8000 ms	1500 ms
Position LFO	stab.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	stab.pos_lfo_beats	0.25 .. 64 beats	16 beats
Cutoff	stab.cutoff	200 .. 18000 Hz	3500 Hz
Resonance	stab.resonance	0 .. 1	0.15
Env Amount	stab.env_amount	0 .. 6 oct	2 oct
Filter Decay	stab.filter_decay	5 .. 3000 ms	200 ms
Key Track	stab.key_track	0 .. 1	0.5
HP Floor	stab.hp_floor	40 .. 400 Hz	250 Hz
HP Track	stab.hp_track	0 .. 1 x f0	1 x f0
Attack	stab.amp_attack	0.3 .. 2000 ms	1 ms

Decay	stab.amp_decay	5 .. 4000 ms	180 ms
Sustain	stab.amp_sustain	0 .. 1	0
Release	stab.amp_release	5 .. 4000 ms	120 ms
Width	stab.width	0 .. 1	0.8
Velocity	stab.vel_sens	0 .. 1	0.4
Delay Send	stab.delay_send	0 .. 1	0.15
Delay Left	stab.delay_left	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Delay Right	stab.delay_right	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/8
Delay Feedback	stab.delay_feedback	0 .. 0.9	0.4
Delay High Pass	stab.delay_high_pass	150 .. 2000 Hz	350 Hz
Delay Low Pass	stab.delay_low_pass	800 .. 16000 Hz	6000 Hz
Room Send	stab.room_send	0 .. 1	0.1
Hall Send	stab.hall_send	0 .. 1	0.25
Duck	stab.duck	0 .. 18 dB	3 dB
Trance Gate	stab.gate	off / on	off
Gate Pattern	stab.gate_pattern	Sixteenths, Eighths, Rolling, Gallop, 3-3-2, Triplets, Trance 1, Trance 2, Trance 3	Trance 1
Gate Depth	stab.gate_depth	0 .. 1	0.85
Gate Duty	stab.gate_duty	0.05 .. 1	0.5
Gate Attack	stab.gate_attack	0.5 .. 60 ms	3 ms
Gate Release	stab.gate_release	0.5 .. 120 ms	12 ms
Gate Tone	stab.gate_tone	0 .. 1	0.4
Level	stab.level	-36 .. 6 dB	-11 dB
Disperse	stab.disperse	0 .. 8 x	0 x
Disperse Freq	stab.disperse_freq	200 .. 8000 Hz	1250 Hz
Drift	stab.drift	0 .. 8 ct	1 ct
Filter Type	stab.filter_type	Low Pass, Band Pass, High Pass, Notch	Low Pass
Glide	stab.glide	0 .. 400 ms	0 ms
Pan	stab.pan	-1 .. 1	0
Oscillator 2	stab.osc2	Off, Supersaw, VA, FM, Wavetable	Off
Osc 2 Mix	stab.osc2_mix	0 .. 1	0

Osc 2 Interval	stab.osc2_interval	-2 Oct, -1 Oct, -5th, Unison, +5th, +1 Oct	-1 Oct
Osc 2 Detune	stab.osc2_detune	-50 .. 50 ct	0 ct
Voice LFO	stab.lfo_beats	0.25 .. 64 beats	8 beats
LFO to Cutoff	stab.lfo_cutoff	0 .. 4 oct	0 oct
LFO to Pitch	stab.lfo_pitch	0 .. 50 ct	0 ct
LFO to Level	stab.lfo_amp	0 .. 1	0
Plate Send	stab.plate_send	0 .. 1	0.15
Slow Movement	stab.slow_mod	0 .. 1	0
Filter Model	stab.filter_model	State Variable, Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder, Korg35 (MS-20), Polivoks, Wasp	State Variable
Filter Mode	stab.filter_mode	0 .. 1	0
Filter Attack	stab.filt_attack	0.1 .. 4000 ms	0.1 ms
Filter Sustain	stab.filt_sustain	0 .. 1	0
Filter Release	stab.filt_release	5 .. 8000 ms	300 ms
Mod Attack	stab.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	stab.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	stab.menv_sustain	0 .. 1	0
Mod Release	stab.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	stab.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	stab.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	stab.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrig	stab.lfo1_retrig	off / on	off
LFO 1 Fade	stab.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	stab.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	stab.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine

LFO 2 Sync	stab.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrig	stab.lfo2_retrig	off / on	off
LFO 2 Fade	stab.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	stab.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	stab.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	stab.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrig	stab.lfo3_retrig	off / on	off
LFO 3 Fade	stab.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	stab.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	stab.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	stab.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrig	stab.lfo4_retrig	off / on	off
LFO 4 Fade	stab.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	stab.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	stab.mx1_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 1 Amount	stab.mx1_amount	-1 .. 1	0
Mod 2 Source	stab.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	stab.mx2_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 2 Amount	stab.mx2_amount	-1 .. 1	0
Mod 3 Source	stab.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	stab.mx3_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 3 Amount	stab.mx3_amount	-1 .. 1	0
Mod 4 Source	stab.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off

Mod 4 Target	stab.mx4_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 4 Amount	stab.mx4_amount	-1 .. 1	0
Mod 5 Source	stab.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	stab.mx5_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 5 Amount	stab.mx5_amount	-1 .. 1	0
Mod 6 Source	stab.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	stab.mx6_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 6 Amount	stab.mx6_amount	-1 .. 1	0
Mod 7 Source	stab.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	stab.mx7_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 7 Amount	stab.mx7_amount	-1 .. 1	0
Mod 8 Source	stab.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	stab.mx8_dst	Off, Pitch, Osc 2 Pitch, Pulse Width, Table Position, FM Index, Cutoff, Resonance, Filter Mode, Level, Pan, Detune	Off
Mod 8 Amount	stab.mx8_amount	-1 .. 1	0

Keys

PARHELION

Uplifting

A

Phrygian

Track

DJ mix

Length

8.0 min

Compose track

New seed

Stop

Mute

+

-

seed 4242

single track, Uplifting Anthem, 137.5 BPM, A Phrygian (BA), 7.4 min, levelling

nothing rerolled

Update check

Full screen

Intro

Groove

Break

Drop

Breakdown

Build

Drop

Outro

1

17

33

49

65

81

97

113

129

145

161

177

193

209

225

241

Set

Arrange

Low End

Drums

Synths

Keys

Orchestra

Effects

Mixer

Perform

Export

Style

PIANO PRESET

<

Brilliant Lauds

>

this track: Brilliant Lauds (Muted Mellow)

INSTRUMENT

Instrument

Baby Grand

Hardness

0.70

Strike Point

0.28

Unison Width

1.21 ct

Inharmonicity

1.13

Impedance

0.55

Stretch

1.13

Condition

0.39 ct

PLAYING

Level

-8.10 dB

Pedal

0.00

Sympathetic

1.00

Phantom

1.00

Damper Noise

0.30

Mechanics

0.30

OUT

Width

0.70

Low Cut

80.0 Hz

Duck

2.00 dB

Room Send

0.05

Plate Send

0.10

Hall Send

0.35

MOD ENVELOPE

Mod Attack

10.0 ms

Mod Decay

800 ms

Mod Sustain

0.00

Mod Release

400 ms

LFO 1

Rate

1.00 Hz

Shape

Sine

Sync

Free

Retrig

Fade

0.00 s

LFO 2

Rate

1.00 Hz

Shape

Sine

Sync

Free

Retrig

Fade

0.00 s

LFO 3

Rate

1.00 Hz

Shape

Sine

Sync

Free

Retrig

Fade

0.00 s

LFO 4

Rate

1.00 Hz

Shape

Sine

Sync

Free

Retrig

Fade

0.00 s

MATRIX 1-2

Mod 1 Source

Off

Mod 1 Target

Off

Mod 1 Amount

0.00

Mod 2 Source

Off

Mod 2 Target

Off

Mod 2 Amount

0.00

MATRIX 3-4

Mod 3 Source

Off

Mod 3 Target

Off

Mod 3 Amount

0.00

Mod 4 Source

Off

Mod 4 Target

Off

Mod 4 Amount

0.00

MATRIX 5-6

Mod 5 Source

Off

Mod 5 Target

Off

Mod 5 Amount

0.00

Mod 6 Source

Off

Mod 6 Target

Off

Mod 6 Amount

0.00

MATRIX 7-8

Mod 7 Source

Off

Mod 7 Target

Off

Mod 7 Amount

0.00

Mod 8 Source

Off

Mod 8 Target

Off

Mod 8 Amount

0.00

CLOUD PRESET

<

Drifting Seed Pearl

>

this track: Drifting Seed Pearl (Crystal Grains)

GRAINS

Level

-60.0 dB

Density

26.3 /s

Grain Size

100 ms

Pitch

0.64

Spray

2.70 s

SOURCES

From Pad

0.80

From Keys

0.50

Plate Send

0.50

The piano is a physical model, not a sample: for every key a string of the length, tension and stiffness of a real instrument (four of them: Grand, Baby Grand, Upright and Soft, the slightly muffled piano of Dream House), one to three strings per note, a felt hammer, a soundboard solved as a ribbed orthotropic plate, the damper, the pedal and the sympathetic strings. Its stretched tuning comes from the strings' stiffness. Its voicing was set note by note against a concert grand (Pianoteq's Steinway D): the level over the keyboard and the velocity, how bright a note is and how much brighter a harder blow makes it, the two stages of its decay, the action's knock and the body in the first moments, how fast the damper takes a note away. In Dream House it plays the motif, the pedal down and lifted at every chord change.

The grain cloud hears the pad, the pluck and the piano and scatters them in grains into the plate -- Deep's haze and Dream House's breath.

PIANO

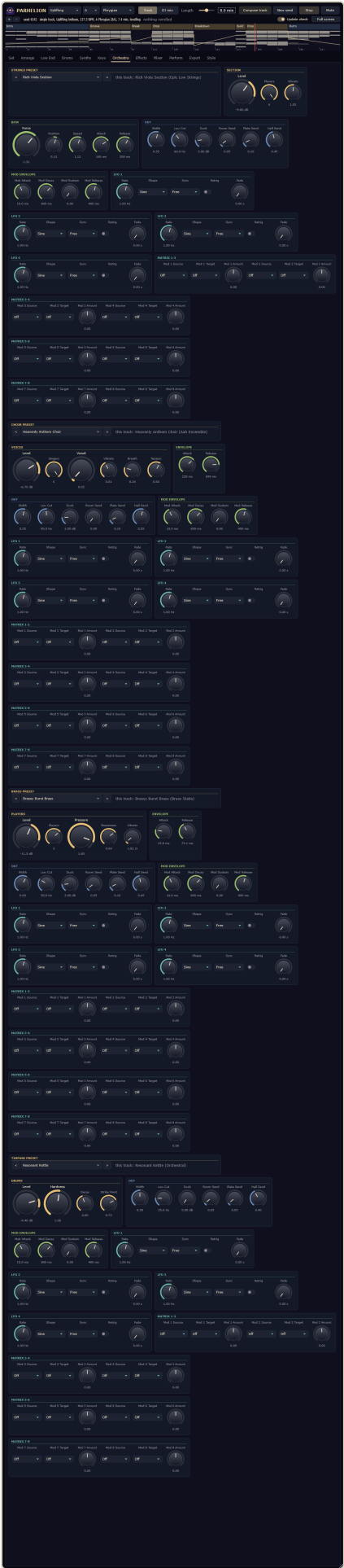
Control	Key	Range	Default
Instrument	<code>piano.instrument</code>	Grand, Baby Grand, Upright, Soft	Grand
Hardness	<code>piano.hardness</code>	0.25 .. 4	1
Strike Point	<code>piano.strike</code>	-1 .. 1	0
Unison Width	<code>piano.unison</code>	0 .. 6 ct	1 ct
Inharmonicity	<code>piano.inharm</code>	0.25 .. 3	1
Impedance	<code>piano.impedance</code>	0.3 .. 3	1
Stretch	<code>piano.stretch</code>	0 .. 2	1
Condition	<code>piano.condition</code>	0 .. 3 ct	0.3 ct
Level	<code>piano.level</code>	-36 .. 6 dB	-6 dB
Pedal	<code>piano.pedal</code>	0 .. 1	0
Sympathetic	<code>piano.sympathetic</code>	0 .. 2	1
Phantom	<code>piano.phantom</code>	0 .. 3	1
Damper Noise	<code>piano.damper_noise</code>	0 .. 1	0.3
Mechanics	<code>piano.mechanics</code>	0 .. 1	0.3
Width	<code>piano.width</code>	0 .. 1	0.7
Low Cut	<code>piano.low_cut</code>	20 .. 400 Hz	80 Hz
Duck	<code>piano.duck</code>	0 .. 18 dB	2 dB
Room Send	<code>piano.room_send</code>	0 .. 1	0.05
Plate Send	<code>piano.plate_send</code>	0 .. 1	0.1
Hall Send	<code>piano.hall_send</code>	0 .. 1	0.35
Mod Attack	<code>piano.menv_attack</code>	0.1 .. 8000 ms	10 ms
Mod Decay	<code>piano.menv_decay</code>	5 .. 12000 ms	800 ms
Mod Sustain	<code>piano.menv_sustain</code>	0 .. 1	0
Mod Release	<code>piano.menv_release</code>	5 .. 12000 ms	400 ms
LFO 1 Rate	<code>piano.lfo1_rate</code>	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	<code>piano.lfo1_shape</code>	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	<code>piano.lfo1_sync</code>	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free

LFO 1 Retrigger	piano.lfo1_retrig	off / on	off
LFO 1 Fade	piano.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	piano.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	piano.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	piano.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	piano.lfo2_retrig	off / on	off
LFO 2 Fade	piano.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	piano.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	piano.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	piano.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	piano.lfo3_retrig	off / on	off
LFO 3 Fade	piano.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	piano.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	piano.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	piano.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	piano.lfo4_retrig	off / on	off
LFO 4 Fade	piano.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	piano.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	piano.mx1_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 1 Amount	piano.mx1_amount	-1 .. 1	0
Mod 2 Source	piano.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	piano.mx2_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 2 Amount	piano.mx2_amount	-1 .. 1	0
Mod 3 Source	piano.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	piano.mx3_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 3 Amount	piano.mx3_amount	-1 .. 1	0
Mod 4 Source	piano.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	piano.mx4_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 4 Amount	piano.mx4_amount	-1 .. 1	0
Mod 5 Source	piano.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	piano.mx5_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 5 Amount	piano.mx5_amount	-1 .. 1	0

Mod 6 Source	piano.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	piano.mx6_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 6 Amount	piano.mx6_amount	-1 .. 1	0
Mod 7 Source	piano.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	piano.mx7_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 7 Amount	piano.mx7_amount	-1 .. 1	0
Mod 8 Source	piano.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	piano.mx8_dst	Off, Pitch, Hardness, Level, Pan	Off
Mod 8 Amount	piano.mx8_amount	-1 .. 1	0

CLOUD

Control	Key	Range	Default
Level	cloud.level	-60 .. 6 dB	-60 dB
Density	cloud.density	0.5 .. 60 /s	10 /s
Grain Size	cloud.size	20 .. 1000 ms	260 ms
Pitch	cloud.pitch	0 .. 1	0.3
Spray	cloud.spray	0.05 .. 4.5 s	2.5 s
From Pad	cloud.pad_send	0 .. 1	0.8
From Keys	cloud.keys_send	0 .. 1	0.5
Plate Send	cloud.plate_send	0 .. 1	0.5



The Cinematic tracks of Uplifting (and some of Dream House and Deep) bring an orchestra, all of it modelled. The strings: six players a note, each a modal string under a bow with the friction of rosin, their own intonation, attack and vibrato, in the body of a violin, viola, cello or double bass by register. The choir: a glottal source with the tension of the voice, five formants per singer on the vowels a, o and u, breath and vibrato. The brass: a lip that buzzes into a bore and its bell, brighter with more breath. The timpani: the modes of a kettle drum under a felt mallet.

The strings enter with the main breakdown's tease and take the melody at its peak; the choir sings at the peak and in the second half of the main drop; at every drop the brass play a braam (the root low, its fifth, the octave) and the timpani strike, after a roll in the two bars before it.

STRINGS

Control	Key	Range	Default
Level	strings.level	-36 .. 6 dB	-6 dB
Players	strings.players	1 .. 6	6
Vibrato	strings.vibrato	0 .. 2	1
Bow Force	strings.pressure	0.3 .. 3	1
Bow Position	strings.position	-1 .. 1	0
Bow Speed	strings.speed	0.3 .. 3	1
Attack	strings.attack	5 .. 800 ms	90 ms
Release	strings.release	10 .. 1500 ms	180 ms
Width	strings.width	0 .. 1	0.55
Low Cut	strings.low_cut	20 .. 400 Hz	60 Hz
Duck	strings.duck	0 .. 18 dB	3 dB
Room Send	strings.room_send	0 .. 1	0.05
Plate Send	strings.plate_send	0 .. 1	0.1
Hall Send	strings.hall_send	0 .. 1	0.45
Mod Attack	strings.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	strings.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	strings.menv_sustain	0 .. 1	0
Mod Release	strings.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	strings.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	strings.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	strings.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	strings.lfo1_retrig	off / on	off
LFO 1 Fade	strings.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	strings.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	strings.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	strings.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	strings.lfo2_retrig	off / on	off
LFO 2 Fade	strings.lfo2_fade	0 .. 8 s	0 s

LFO 3 Rate	strings.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	strings.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	strings.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	strings.lfo3_retrig	off / on	off
LFO 3 Fade	strings.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	strings.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	strings.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	strings.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	strings.lfo4_retrig	off / on	off
LFO 4 Fade	strings.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	strings.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	strings.mx1_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off
Mod 1 Amount	strings.mx1_amount	-1 .. 1	0
Mod 2 Source	strings.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	strings.mx2_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off
Mod 2 Amount	strings.mx2_amount	-1 .. 1	0
Mod 3 Source	strings.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	strings.mx3_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off
Mod 3 Amount	strings.mx3_amount	-1 .. 1	0
Mod 4 Source	strings.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	strings.mx4_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off
Mod 4 Amount	strings.mx4_amount	-1 .. 1	0
Mod 5 Source	strings.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	strings.mx5_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off
Mod 5 Amount	strings.mx5_amount	-1 .. 1	0
Mod 6 Source	strings.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	strings.mx6_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off

Mod 6 Amount	strings.mx6_amount	-1 .. 1	0
Mod 7 Source	strings.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	strings.mx7_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off
Mod 7 Amount	strings.mx7_amount	-1 .. 1	0
Mod 8 Source	strings.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	strings.mx8_dst	Off, Pitch, Bow Pressure, Bow Speed, Bow Position, Vibrato, Vibrato Rate, Level, Pan	Off
Mod 8 Amount	strings.mx8_amount	-1 .. 1	0

CHOIR

Control	Key	Range	Default
Level	choir.level	-36 .. 6 dB	-6 dB
Singers	choir.singers	1 .. 6	6
Vowel	choir.vowel	0 .. 1	0
Vibrato	choir.vibrato	0 .. 2	1
Breath	choir.breath	0 .. 1	0.25
Tension	choir.tension	0 .. 1	0.4
Attack	choir.attack	5 .. 1500 ms	250 ms
Release	choir.release	10 .. 2000 ms	400 ms
Width	choir.width	0 .. 1	0.55
Low Cut	choir.low_cut	20 .. 400 Hz	90 Hz
Duck	choir.duck	0 .. 18 dB	3 dB
Room Send	choir.room_send	0 .. 1	0
Plate Send	choir.plate_send	0 .. 1	0.1
Hall Send	choir.hall_send	0 .. 1	0.55
Mod Attack	choir.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	choir.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	choir.menv_sustain	0 .. 1	0
Mod Release	choir.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	choir.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	choir.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	choir.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	choir.lfo1_retrig	off / on	off
LFO 1 Fade	choir.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	choir.lfo2_rate	0.01 .. 40 Hz	1 Hz

LFO 2 Shape	choir.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	choir.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	choir.lfo2_retrigger	off / on	off
LFO 2 Fade	choir.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	choir.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	choir.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	choir.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	choir.lfo3_retrigger	off / on	off
LFO 3 Fade	choir.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	choir.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	choir.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	choir.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	choir.lfo4_retrigger	off / on	off
LFO 4 Fade	choir.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	choir.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	choir.mx1_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off
Mod 1 Amount	choir.mx1_amount	-1 .. 1	0
Mod 2 Source	choir.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	choir.mx2_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off
Mod 2 Amount	choir.mx2_amount	-1 .. 1	0
Mod 3 Source	choir.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	choir.mx3_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off
Mod 3 Amount	choir.mx3_amount	-1 .. 1	0
Mod 4 Source	choir.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	choir.mx4_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off
Mod 4 Amount	choir.mx4_amount	-1 .. 1	0
Mod 5 Source	choir.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	choir.mx5_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off
Mod 5 Amount	choir.mx5_amount	-1 .. 1	0
Mod 6 Source	choir.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	choir.mx6_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off

Mod 6 Amount	choir.mx6_amount	-1 .. 1	0
Mod 7 Source	choir.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	choir.mx7_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off
Mod 7 Amount	choir.mx7_amount	-1 .. 1	0
Mod 8 Source	choir.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	choir.mx8_dst	Off, Pitch, Vowel, Tension, Breath, Vibrato, Formant, Level, Pan	Off
Mod 8 Amount	choir.mx8_amount	-1 .. 1	0

BRASS

Control	Key	Range	Default
Level	brass.level	-36 .. 6 dB	-6 dB
Players	brass.players	1 .. 4	3
Pressure	brass.pressure	0.3 .. 2	1
Brassiness	brass.brassiness	0 .. 1	0.35
Attack	brass.attack	5 .. 800 ms	40 ms
Release	brass.release	10 .. 1500 ms	150 ms
Vibrato	brass.vibrato	0 .. 40 ct	6 ct
Width	brass.width	0 .. 1	0.6
Low Cut	brass.low_cut	20 .. 400 Hz	50 Hz
Duck	brass.duck	0 .. 18 dB	3 dB
Room Send	brass.room_send	0 .. 1	0.05
Plate Send	brass.plate_send	0 .. 1	0.1
Hall Send	brass.hall_send	0 .. 1	0.45
Mod Attack	brass.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	brass.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	brass.menv_sustain	0 .. 1	0
Mod Release	brass.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	brass.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	brass.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	brass.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	brass.lfo1_retrig	off / on	off
LFO 1 Fade	brass.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	brass.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	brass.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	brass.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	brass.lfo2_retrig	off / on	off

LFO 2 Fade	brass.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	brass.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	brass.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	brass.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	brass.lfo3_retrigger	off / on	off
LFO 3 Fade	brass.lfo3_fade	0 .. 8 s	0 s
LFO 4 Rate	brass.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	brass.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	brass.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	brass.lfo4_retrigger	off / on	off
LFO 4 Fade	brass.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	brass.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	brass.mx1_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 1 Amount	brass.mx1_amount	-1 .. 1	0
Mod 2 Source	brass.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	brass.mx2_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 2 Amount	brass.mx2_amount	-1 .. 1	0
Mod 3 Source	brass.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	brass.mx3_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 3 Amount	brass.mx3_amount	-1 .. 1	0
Mod 4 Source	brass.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	brass.mx4_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 4 Amount	brass.mx4_amount	-1 .. 1	0
Mod 5 Source	brass.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	brass.mx5_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 5 Amount	brass.mx5_amount	-1 .. 1	0
Mod 6 Source	brass.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	brass.mx6_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 6 Amount	brass.mx6_amount	-1 .. 1	0
Mod 7 Source	brass.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off

Mod 7 Target	brass.mx7_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 7 Amount	brass.mx7_amount	-1 .. 1	0
Mod 8 Source	brass.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	brass.mx8_dst	Off, Pitch, Breath, Brassiness, Vibrato, Level, Pan	Off
Mod 8 Amount	brass.mx8_amount	-1 .. 1	0

TIMPANI

Control	Key	Range	Default
Level	timpani.level	-36 .. 6 dB	-6 dB
Hardness	timpani.hardness	0.25 .. 4	1
Decay	timpani.decay	0.3 .. 3	1
Strike Point	timpani.strike	0.3 .. 0.9	0.68
Width	timpani.width	0 .. 1	0.5
Low Cut	timpani.low_cut	20 .. 400 Hz	35 Hz
Duck	timpani.duck	0 .. 18 dB	0 dB
Room Send	timpani.room_send	0 .. 1	0.05
Plate Send	timpani.plate_send	0 .. 1	0.05
Hall Send	timpani.hall_send	0 .. 1	0.4
Mod Attack	timpani.menv_attack	0.1 .. 8000 ms	10 ms
Mod Decay	timpani.menv_decay	5 .. 12000 ms	800 ms
Mod Sustain	timpani.menv_sustain	0 .. 1	0
Mod Release	timpani.menv_release	5 .. 12000 ms	400 ms
LFO 1 Rate	timpani.lfo1_rate	0.01 .. 40 Hz	1 Hz
LFO 1 Shape	timpani.lfo1_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 1 Sync	timpani.lfo1_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 1 Retrigger	timpani.lfo1_retrig	off / on	off
LFO 1 Fade	timpani.lfo1_fade	0 .. 8 s	0 s
LFO 2 Rate	timpani.lfo2_rate	0.01 .. 40 Hz	1 Hz
LFO 2 Shape	timpani.lfo2_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 2 Sync	timpani.lfo2_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 2 Retrigger	timpani.lfo2_retrig	off / on	off
LFO 2 Fade	timpani.lfo2_fade	0 .. 8 s	0 s
LFO 3 Rate	timpani.lfo3_rate	0.01 .. 40 Hz	1 Hz
LFO 3 Shape	timpani.lfo3_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 3 Sync	timpani.lfo3_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 3 Retrigger	timpani.lfo3_retrig	off / on	off
LFO 3 Fade	timpani.lfo3_fade	0 .. 8 s	0 s

LFO 4 Rate	timpani.lfo4_rate	0.01 .. 40 Hz	1 Hz
LFO 4 Shape	timpani.lfo4_shape	Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random	Sine
LFO 4 Sync	timpani.lfo4_sync	Free, 4 Bars, 2 Bars, 1 Bar, 1/2, 1/4, 1/8, 1/16, 1/4 T, 1/8 T	Free
LFO 4 Retrigger	timpani.lfo4_retrigger	off / on	off
LFO 4 Fade	timpani.lfo4_fade	0 .. 8 s	0 s
Mod 1 Source	timpani.mx1_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 1 Target	timpani.mx1_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off
Mod 1 Amount	timpani.mx1_amount	-1 .. 1	0
Mod 2 Source	timpani.mx2_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 2 Target	timpani.mx2_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off
Mod 2 Amount	timpani.mx2_amount	-1 .. 1	0
Mod 3 Source	timpani.mx3_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 3 Target	timpani.mx3_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off
Mod 3 Amount	timpani.mx3_amount	-1 .. 1	0
Mod 4 Source	timpani.mx4_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 4 Target	timpani.mx4_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off
Mod 4 Amount	timpani.mx4_amount	-1 .. 1	0
Mod 5 Source	timpani.mx5_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 5 Target	timpani.mx5_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off
Mod 5 Amount	timpani.mx5_amount	-1 .. 1	0
Mod 6 Source	timpani.mx6_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 6 Target	timpani.mx6_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off
Mod 6 Amount	timpani.mx6_amount	-1 .. 1	0
Mod 7 Source	timpani.mx7_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 7 Target	timpani.mx7_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off
Mod 7 Amount	timpani.mx7_amount	-1 .. 1	0
Mod 8 Source	timpani.mx8_src	Off, LFO 1, LFO 2, LFO 3, LFO 4, Mod Env, Filter Env, Velocity, Key, Random, Wheel, Pressure, Energy	Off
Mod 8 Target	timpani.mx8_dst	Off, Pitch, Hardness, Strike, Decay, Level, Pan	Off

Mod 8	timpani.mx8_amount	-1 .. 1	0
Amount			

Effects



The effects are synthesised as well: the riser and the sweep of the build, the reverse crash and the impact on the drop, the downlifter after it, the sub drop, swells, zaps and atmospheres, each from its own group of presets.

The sends: a room, a plate and a long hall, which every voice sends into by its own knobs (the hats into the room and the percussion into the plate by the knobs here), each with a low cut at its input and its return ducked by the pump. The composer draws the hall long in a breakdown and short in a drop.

SFX

Control	Key	Range	Default
Level	<code>sfx.level</code>	-36 .. 6 dB	-3 dB
Noise	<code>sfx.noise</code>	0 .. 1	0.6
Resonance	<code>sfx.resonance</code>	0 .. 1	0.5
Brightness	<code>sfx.brightness</code>	0 .. 1	0.5
Impact Decay	<code>sfx.impact_decay</code>	200 .. 4000 ms	1400 ms
Vowel	<code>sfx.vowel</code>	0 .. 1	0.3

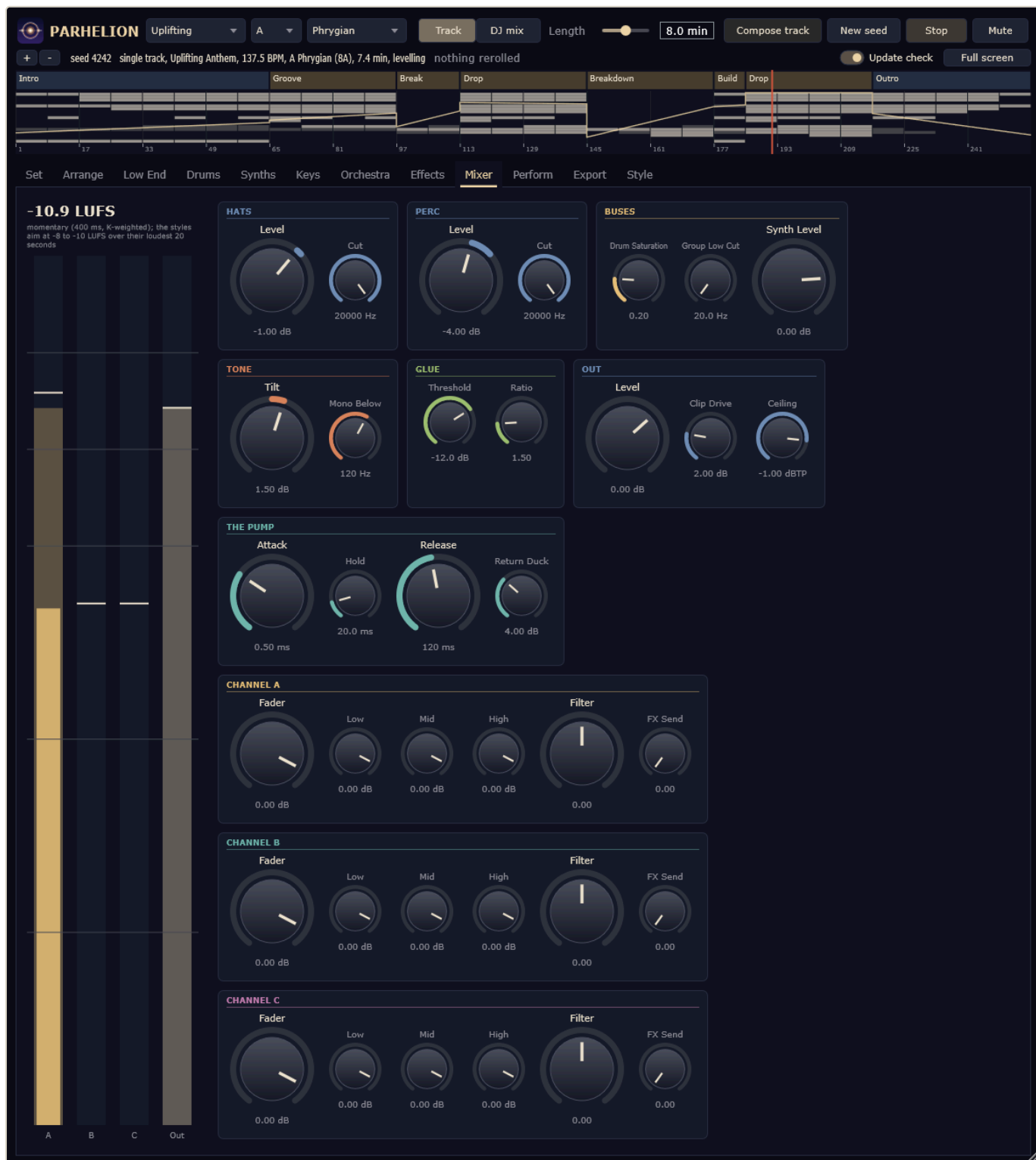
Swell Decay	sfx.swell_decay	200 .. 6000 ms	1500 ms
Width	sfx.width	0 .. 1	0.7
Room Send	sfx.room_send	0 .. 1	0
Hall Send	sfx.hall_send	0 .. 1	0.35
Duck	sfx.duck	0 .. 18 dB	2 dB
Sub Drop	sfx.sub_level	-36 .. 6 dB	-14 dB
Sub Duck	sfx.sub_duck	0 .. 1	1
Wander	sfx.wander	off / on	off
Wander Send	sfx.wander_send	0 .. 1	0.85
Riser	sfx.preset_riser	0 .. 256	0
Downlifter	sfx.preset_downlifter	0 .. 128	0
Impact	sfx.preset_impact	0 .. 128	0
Sweep	sfx.preset_sweep	0 .. 256	0
Formant Shot	sfx.preset_formant_shot	0 .. 96	0
Reverse Swell	sfx.preset_reverse_swell	0 .. 256	0
Zap	sfx.preset_zap	0 .. 96	0
Squelch	sfx.preset_squelch	0 .. 128	0
Bubble	sfx.preset_bubble	0 .. 64	0
Reverse Crash	sfx.preset_reverse_crash	0 .. 128	0
Atmosphere	sfx.preset_atmosphere	0 .. 512	0
Plate Send	sfx.plate_send	0 .. 1	0.1

SENDS

Control	Key	Range	Default
Room Size	sends.room_size	0.3 .. 3	0.7
Room Decay	sends.room_decay	0.2 .. 4 s	1 s
Room Return	sends.room_return	-60 .. 18 dB	6 dB
Plate Decay	sends.plate_decay	0.3 .. 6 s	1.5 s
Plate Damping	sends.plate_damping	0 .. 1	0.4
Plate Pre-Delay	sends.plate_predelay	0 .. 100 ms	20 ms
Plate Return	sends.plate_return	-60 .. 18 dB	0 dB
Hall Size	sends.hall_size	0.5 .. 3	2.2
Hall Decay	sends.hall_decay	0.5 .. 14 s	6 s
Hall Damping	sends.hall_damping	0 .. 1	0.45
Hall Pre-Delay	sends.hall_predelay	0 .. 120 ms	60 ms
Hall Return	sends.hall_return	-60 .. 18 dB	0 dB
Send Low Cut	sends.low_cut	80 .. 800 Hz	280 Hz
Return High Cut	sends.high_cut	2000 .. 18000 Hz	9000 Hz
Hats to Room	sends.hats_send	0 .. 1	0.15

Perc to Plate	sends.perc_send	0 .. 1	0.2
---------------	-----------------	--------	-----

Mixer and Master



The meters: each deck after its mixer channel, the output, and the loudness of the last 400 ms. Under them the mix buses (the Drums chapter), the pump (the Low End chapter) and the master: the glue compressor, a tilt of the highs against the lows, the side under Mono Below made mono, the level, a clipper at four times the rate and a true-peak limiter at the ceiling.

A deck's channel is a DJ mixer's: the fader, a three-band isolator (200 Hz and 2.5 kHz, each band from a kill to +6 dB), a filter that is a low pass to the left and a high pass to the right, and the send into the mixer's effects. In a set the composer moves them as a DJ would; in the plugin they are yours too.

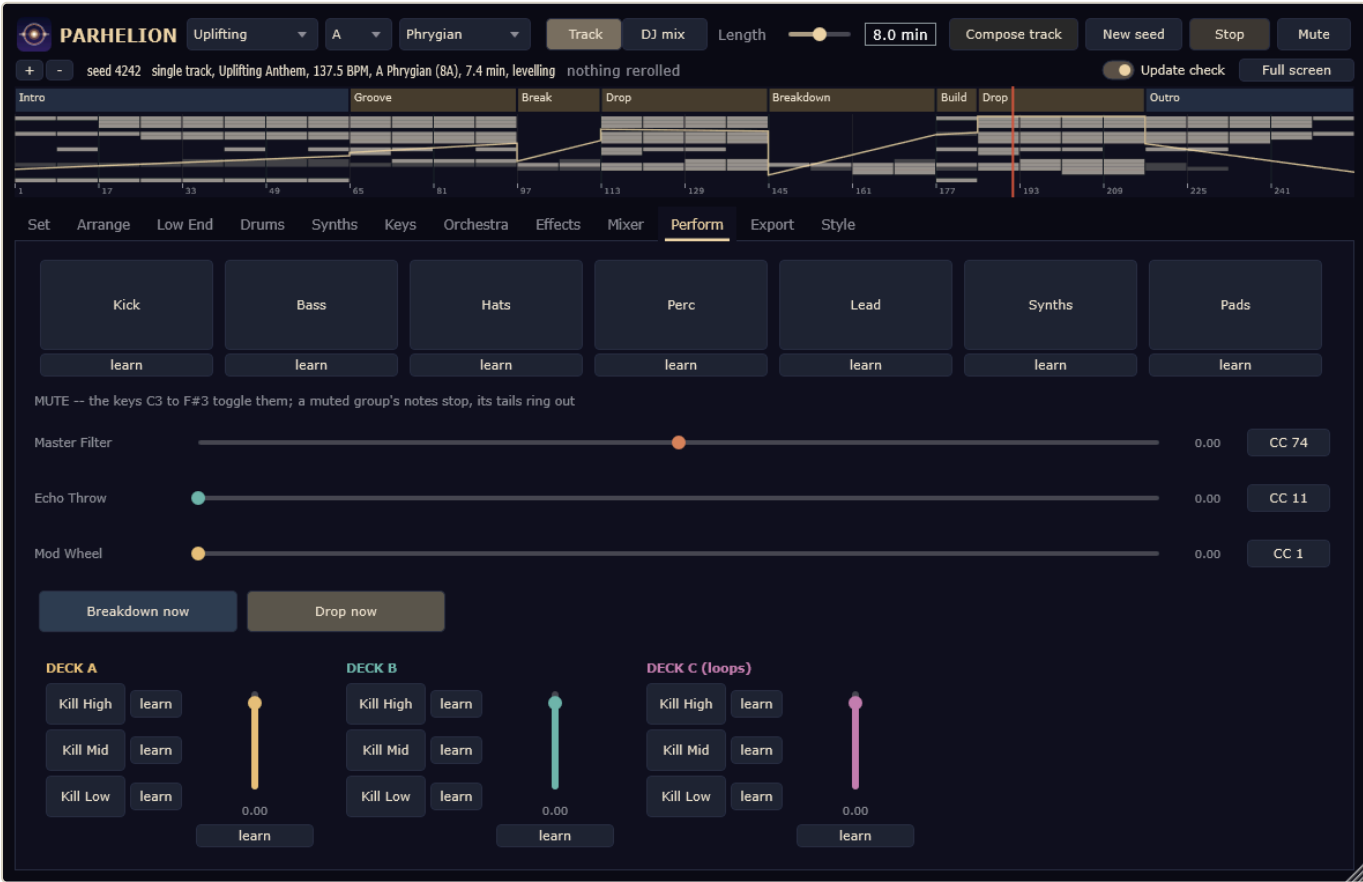
MASTER

Control	Key	Range	Default
Level	master.level	-24 .. 12 dB	0 dB
Glue Threshold	master.threshold	-40 .. 0 dB	-12 dB
Glue Ratio	master.ratio	1 .. 10	1.5
Clip Drive	master.clip	0 .. 9 dB	2 dB
Ceiling	master.ceiling	-6 .. 0 dBTP	-1 dBTP
Mono Below	master.mono_below	40 .. 250 Hz	120 Hz
Tilt	master.tilt	-12 .. 12 dB	1.5 dB

DECK (3 INSTANCES)

Control	Key	Range	Default
Fader	deckN.fader	-60 .. 6 dB	0 dB
Low	deckN.low	-60 .. 6 dB	0 dB
Mid	deckN.mid	-60 .. 6 dB	0 dB
High	deckN.high	-60 .. 6 dB	0 dB
Filter	deckN.filter	-1 .. 1	0
FX Send	deckN.fx_send	0 .. 1	0

Perform



As at a mixer. Seven mutes -- kick, bass, hats, percussion, lead, the other synths, pads --: a muted group's notes stop and its tails ring out. The master filter takes the whole mix, the echo throw sends it into the mixer's echo, and the mod wheel drives every voice's modulation matrix where its preset listens to it. Each deck has its kills and its fader.

Breakdown now and Drop now rewrite the playing track from its next 8-bar line (or the one after it, where that is less than six seconds away): into a breakdown with its build and a drop after it, or into the drop at once, after a beat of silence. Everything before that line stays as it was, and the track plays on without a gap. A second one keeps the first: the track is rewritten from both lines. A set is not rewritten -- its next blend would have to move with it.

Every control can be learned from a MIDI controller: click learn, move the controller (a right click forgets it). By default controller 74 moves the filter, controller 11 the throw and the mod wheel the wheel; channel pressure is the matrices' pressure; the keys from middle C up (C to F#) toggle the seven mutes. The bindings are saved with the project.

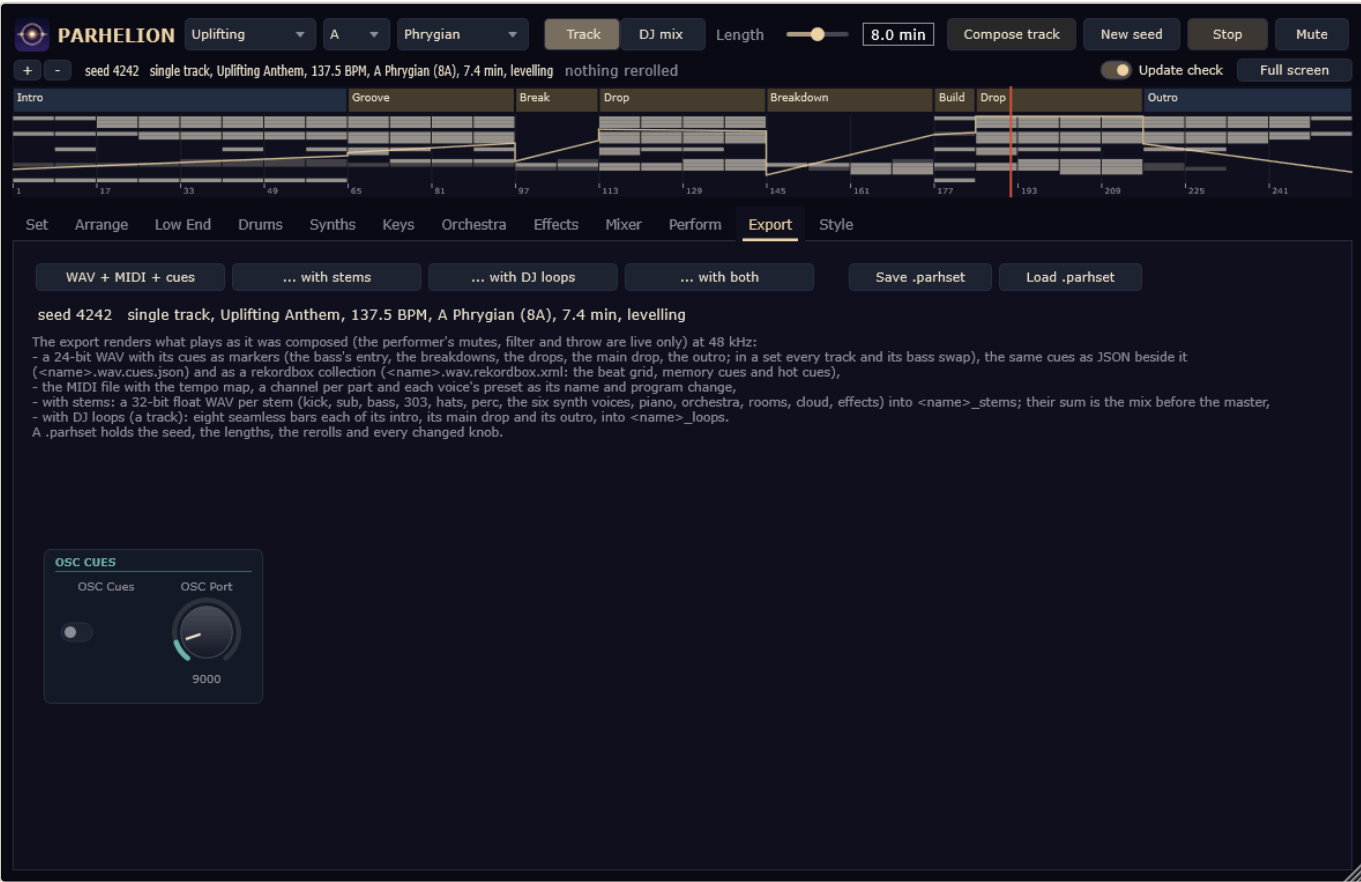
The perform controls are live only: an export or parh_render plays the track as it was composed.

PERFORM

Control	Key	Range	Default
Master Filter	perform.filter	-1 .. 1	0
Echo Throw	perform.throw	0 .. 1	0
Mute Kick	perform.mute_kick	off / on	off

Mute Bass	perform.mute_bass	off / on	off
Mute Hats	perform.mute_hats	off / on	off
Mute Perc	perform.mute_perc	off / on	off
Mute Lead	perform.mute_lead	off / on	off
Mute Synths	perform.mute_synths	off / on	off
Mute Pads	perform.mute_pads	off / on	off
Mod Wheel	perform.wheel	0 .. 1	0
Pressure	perform.pressure	0 .. 1	0

Export and cues



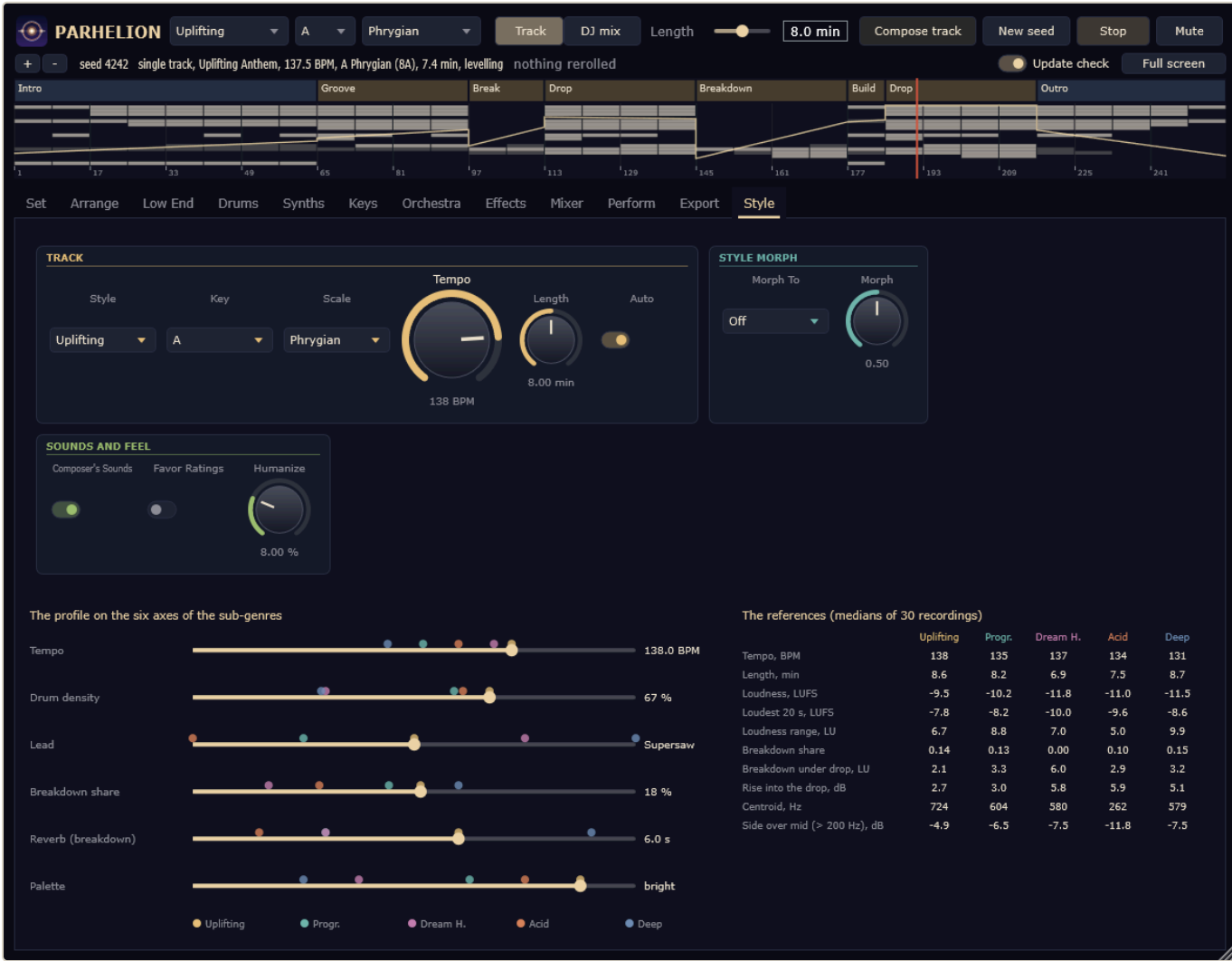
The export renders what plays at 48 kHz: a 24-bit WAV with cue markers (the bass's entry, the breakdowns, the drops, the main drop, the outro; in a set every track and its bass swap) and the same cues as JSON beside it (<name>.wav.cues.json), for DJ software that does not read the WAV's markers, and a rekordbox collection beside it (<name>.wav.rekordbox.xml: the beat grid, following a set's tempo bar by bar, every cue a memory cue and the first eight hot cues; import it in rekordbox under Preferences > Advanced > rekordbox xml); the MIDI file with the tempo map, a channel per part and each voice's preset as its name and program change; on request the stems -- a WAV per part -- and the DJ loops of a track: eight seamless bars each of its intro, its main drop and its outro. A .parhset file holds the seed, the lengths, the rerolls and every changed knob.

The OSC cues tell a visualiser what the score knows, at the moment it is heard: the beat and the tempo, the bar, the section's name, what the layer matrix changes, the key. They go to this machine at the port given, or to PARH_CUE_HOST.

```
/parh/beat i f    beat number, tempo
/parh/bar i      bar number
/parh/block s    the section's name ("Breakdown", "Drop", a set's track)
/parh/op s       the layer matrix's change ("add lead", "filter pluck", "remove kick")
/parh/key s      the key as a Camelot label ("8A")
```

CUE

Control	Key	Range	Default
OSC Cues	cue.enabled	off / on	off
OSC Port	cue.port	1024 .. 65535	9000



Where the knobs' profile stands among the five styles, on six axes: the tempo, the density of the drums, how much the lead carries, the share of the breakdowns, the length of the breakdown's reverb and the brightness of the palette the composer chooses its presets from. The markers are the five profiles; the bar is the profile of the knobs, with its morph. Under them the references' medians per style, as they were measured: tempo, length, integrated loudness and the loudest 20 seconds, the loudness range, the breakdown's share and its distance to the drop, the rise into the drop, the spectral centroid and the width above 200 Hz.

Meta Quest

The whole generator runs on the headset (build-quest\ParhelionQuest.apk, see Quest/README.md). The hands are the hands at the mixer: left pinch play and stop, right pinch kick out and in, the right pinch held Breakdown now or Drop now (handed over on a beat, without a gap: a second engine takes over from the first), both hands together the next track; the left hand's height the master filter, the right's the echo throw. Above and ahead of the player the parhelion hangs in the sky at the size of a room: the sun swells with the kick, the halo is the bar with the parts' notes on it, the sun dogs burn with the track's energy, and the line through them is the track with its sections. A file parh.cfg in the app's folder sets the seed, the style, a set's length and any knob. The headset's quality plays fewer unison oscillators, voices and players and solves the filter models more lightly.

The command line

parh_render composes and renders without a window, the same samples the plugin plays:

```
parh_render --seed 7 --out track.wav --midi track.mid --stems stems --loops loops
parh_render --style Acid --seed 3 --minutes 7 --out acid.wav --plan-json acid.json
parh_render --dj 120 --seed 7 --set "set.dramaturgy=Sunrise; set.journey=Wander" --out set.wav
parh_render --seed 7 --reroll motif --out other_hook.wav
parh_render --list                    every parameter with its value
parh_render --bench --quality quest how fast, at the headset's quality
```

Tools/calibrate.py renders every style and measures its tracks exactly as the references were measured; the evaluation (Tools/eval_report.py) adds the section boundaries it hears against the planned ones, the key it hears against the planned one, and the rise into each drop.